

Covalent Organic Frameworks for Photocatalytic Hydrogen Evolution: Linkage Chemistry, Band-Structure Engineering, and the Path to Scalable Solar Fuels

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Abstract

Covalent organic frameworks (COFs) have moved from a crystallographic curiosity to one of the most designable classes of organic photocatalysts for the hydrogen evolution reaction (HER). Yet reported activities span four orders of magnitude, and it is rarely obvious which of the many co-varying structural choices deserves the credit. This Review organizes the evidence along a single causal chain – linkage chemistry sets conjugation and chemical stability; conjugation and stacking set the band structure; band edges and exciton binding govern charge separation; and interfacial engineering with cocatalysts converts separated charges into hydrogen – rather than by chronology. We first map the four dominant linkage families (β -ketoenamine, imine, vinylene/ sp^2 -carbon, and triazine) onto this chain, then survey the design levers that act on each link: donor–acceptor backbone engineering, linkage protonation, single-atom and low-loading cocatalysts, and Z/S-scheme heterojunctions. We treat mechanistic spectroscopy as an inference chain and collect matched-protocol performance comparisons with explicit irradiation, donor, and normalization variables, including apparent quantum yields up to $\sim 80\%$ at 420 nm and rates approaching $400 \text{ mmol g}^{-1} \text{ h}^{-1}$ in the best-reported systems. As a falsifiable synthesis of the design rules, we outline a pre-registered improvement route for the workhorse β -ketoenamine COF TpPa-1, with quantitative gates covering structure retention, mechanistic gain, activity, and durability. We close with the measurement, stability, and reactor-scale questions that separate laboratory quantum efficiency from scalable solar fuel production.

1 Introduction

Solar hydrogen is attractive because it stores intermittent sunlight in a dispatchable chemical bond, yet particulate photocatalysis remains limited by a chain of losses – photon harvesting, exciton dissociation, charge extraction, and surface turnover – each of which can silently consume an order of magnitude in efficiency [28, 62, 94]. Early oxide studies established that carrier lifetime, not absorption, is often the controlling variable: hole lifetimes in TiO_2 correlate directly with water-splitting ability, and oxygen quantum yields depend strongly on light intensity [94]. Any candidate material class therefore has to be judged not on a single headline rate but on how well its structure can be engineered along every link of that chain [12, 26].

Covalent organic frameworks (COFs) enter this competition with an unusual proposition: fully organic, crystalline, porous semiconductors whose band structure, exciton physics, and active-site placement are all encoded in molecular building blocks [60, 68, 80, 87]. The reticular toolbox unites the tunability of molecular photochemistry with the robustness and long-range order of extended solids [3, 30, 49], and two-dimensional (2D) COFs in particular stack conjugated sheets into columnar π -arrays that support exciton migration and out-of-plane charge transport [47, 96, 140]. Since the first demonstration that a hydrazone-linked COF evolves hydrogen under visible light, the field has expanded through azine, imine, β -ketoenamine, triazine, and vinylene linkage generations [32, 77, 86, 112], with recent systems reporting apparent quantum efficiencies (AQE) above 80% at 450 nm [51] and mass-normalized rates approaching $400 \text{ mmol g}^{-1} \text{ h}^{-1}$ [23,

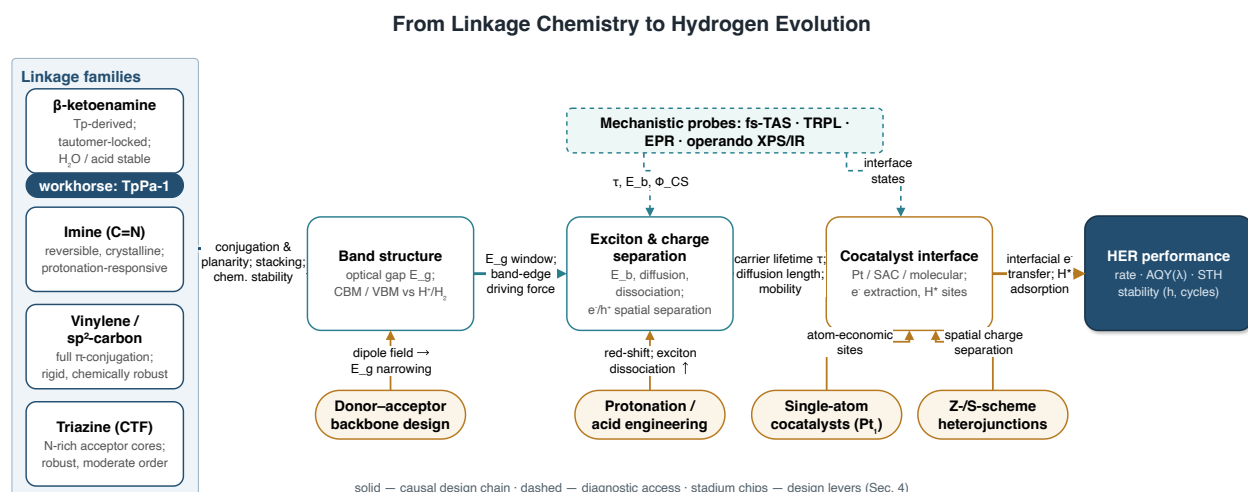


Figure 1: The causal design chain that organizes this Review. Linkage chemistry (left) sets conjugation, planarity, and chemical stability; these determine the band structure; band edges and exciton binding govern charge separation; and the cocatalyst interface converts extracted electrons into hydrogen. Solid arrows denote the causal chain, stadium chips denote the design levers of Section 2, and the dashed channel marks the mechanistic probes of Section 3.

136]. Designability, however, does not by itself establish efficient charge separation or scalable operation: strongly bound Frenkel excitons, layer disorder, and cocatalyst dependence remain recurring bottlenecks [4, 85, 128].

The literature now contains hundreds of COF photocatalysts but comparatively few controlled comparisons, and reviews that organize the material by publication date tend to obscure which structural lever actually moved which observable [1, 116, 120, 142]. This Review therefore organizes evidence by the causal design chain summarized in Figure 1: linkage chemistry $\rightarrow$ conjugation and planarity $\rightarrow$ band gap and band-edge positions $\rightarrow$ exciton dissociation and charge separation $\rightarrow$ cocatalyst interface $\rightarrow$ HER performance. Each design lever – donor–acceptor (D–A) backbone engineering, linkage protonation, single-atom cocatalysts, and Z/S-scheme heterojunctions – is treated as an intervention on a specific link, and mechanistic spectroscopy is treated as the measurement that can distinguish competing explanations [12, 26, 89].

The remainder of the Review follows this chain. Section 2 maps the linkage families and the design levers acting on band structure and exciton physics. Section 3 reviews the mechanistic characterization toolbox, from femtosecond transient absorption to operando spectroscopy. Section 4 examines synthesis methodology and crystallinity control, the practical variables that gate every other claim. Section 5 compares reported performance under explicit protocol variables. Section 6 converts the accumulated design rules into a pre-registered, falsifiable improvement route for the workhorse β -ketoenamine COF TpPa-1. Section 7 confronts the stability, measurement, and reactor questions that stand between laboratory AQE and solar fuel production, and Section 8 concludes.

2 From Linkage Chemistry to Band Structure: The Design Levers

2.1 Four linkage families, four positions on the stability–conjugation frontier

The linkage is the single decision that most strongly constrains everything downstream, because it fixes both the degree of π -delocalization across the framework and the hydrolytic stability that photocatalysis in water demands [30, 67, 80, 87]; linkage-first taxonomies now organize COF photocatalysis across reactions well beyond HER [115]. Table 1 summarizes the four families that dominate HER work.

β -Ketoenamine (Tp-based) COFs, formed from 1,3,5-triformylphloroglucinol (Tp) and diamines

Table 1: The four dominant linkage families for COF photocatalytic HER. Rates depend strongly on protocol (Section 5); exemplars are indicative, not rankings.

Family	Formation	Conjugation / planarity	Stability	HER exemplar
β -Ketoenamine	Tp + diamine; tautomer-locked	Interrupted in-plane π ; ESIPT-active	Boiling water, strong acid	TpPa-1 and Tp-COF composites [97, 112]
Imine (C=N)	Reversible Schiff base	Moderate; protonation-responsive	Good; acid-sensitive linkage	Protonated D–A imine COFs, 20.7 mmol g ⁻¹ h ⁻¹ [114]
Vinylene / sp ² c	Aldol / Knoevenagel / HWE	Full π -conjugation; rigid	9 M HCl / 9 M NaOH	DBC-sp ² c-COF, 105 mmol g ⁻¹ h ⁻¹ (saltwater) [8, 122]
Triazine (CTF-like)	Nitrile cyclization; polycondensation	Planar N-rich acceptor cores	Robust; moderate order	TP-COF / CTF heterostructures [33, 110]

through combined Schiff-base condensation and irreversible enol-to-keto tautomerization, trade some in-plane conjugation for exceptional acid and water stability, which is why this family – with TpPa-1 as its workhorse – accounts for the largest share of reported COF photocatalysts and composites [92, 97, 112]. The keto-locked backbone also supports excited-state intramolecular proton transfer (ESIPT), whose induced photoisomerization must be considered when assigning activity differences within the family [105]. Imine (C=N) COFs retain reversible linkage formation and therefore high crystallinity, and their lone-pair-rich linkages are protonation-responsive, a property Section 2.3 develops into a full design lever [5, 59, 114]. Vinylene and sp²-carbon frameworks extend conjugation through C=C bonds that cannot hydrolyze, combining full π -delocalization with chemical robustness: sp²-carbon COFs manage the complete photon-to-charge event sequence within one lattice [47], survive 9 M HCl and 9 M NaOH [8], and, in a matched comparison, an sp²-carbon-linked COF catalyzed overall water splitting with an AQE of 2.53% at 420 nm where its imine-linked analogue was inactive [11]. Kagome-lattice sp²c-COFs templated from an imine precursor reach 105.3 and 80.0 mmol g⁻¹ h⁻¹ in 3 wt% saltwater and natural seawater, respectively [122], and pyrene-based sp²c frameworks outperform their imine counterparts for the same topology [34, 58]. Extending the acetylenic logic, a graphdiyne-like diacetylene-rich linker delivers 10.16 mmol g⁻¹ h⁻¹ against 3.71 and 1.13 for its non-GDY analogues [53]. Aldol, Knoevenagel, and Horner–Wadsworth–Emmons condensations now give synthetic access across this family [77]. Finally, triazine-based frameworks contribute nitrogen-rich acceptor cores with planar, fully conjugated attributes shared with g-C₃N₄ and CTFs [33]; base- and temperature-controlled routes now deliver crystalline triazine-sp²c hybrids [125], and the natural affinity of triazine cores for g-C₃N₄ interfaces carries over to junction chemistry beyond HER [78].

Within each family, linkage-level editing tunes activity further: cyano versus unsubstituted vinylene isomers show distinct structure–activity relationships [70], single-atom skeletal editing opens irreversible Katritzky-type routes to acridinium frameworks [76], and a matched three-linkage platform (imine, enamine, and alkene on the same knot–linker pair) isolates the linkage contribution directly, with the alkene member reaching an AQE of 6.7% at 420 nm [69]. The emerging design rule is that conjugation and hydrolytic robustness need not be traded off – vinylene chemistry achieves both – but crystallinity cost and synthetic accessibility still favor imine and β -ketoenamine platforms for systematic studies [77, 106].

2.2 Donor–acceptor backbone engineering

The controlled variable in donor–acceptor (D–A) design is the intramolecular dipole field, and its expected consequence is twofold: a narrowed optical gap and a driving force that pre-separates the Frenkel exciton along the donor-to-acceptor axis [55, 56, 89]. Computation now leads synthesis here: density-functional screening of triazine- and benzene-cored acceptor series shows exciton binding energy (E_b) varies systematically with donor strength, and the synthesized activity order follows the calculated E_b ranking [81]. The benchmark experimental results are consistent: a β -ketene–cyano D–A pair prolongs carrier lifetime dramatically and delivers an AQE of 82.6% at 450 nm with Pt [51]; a pyrene-donor/dual-benzothiadiazole-acceptor COF averages $12.7 \text{ mmol g}^{-1} \text{ h}^{-1}$ under visible light, among the highest for D–A-type systems [55]; and trithiophene-based D–A frameworks operate cocatalyst-free [71].

Three refinements sharpen the basic rule. First, symmetry matters as much as strength: highly symmetric D–A architectures cancel their dipoles and weaken the built-in electric field, so breaking symmetry – excitonic dipole reorientation [4], asymmetric imine-linker orientation (raising solar-to-hydrogen efficiency from 2.76% to 4.24% [59]), or ground-state charge-transfer bonding that directly intensifies the built-in field [91] – recovers the driving force. Multistep charge transfer in single-crystalline COFs shows how far directional field design can be pushed [48]. Second, the linker length and core conjugation co-tune the gap: biphenyl linkers outperform benzene and triphenyl analogues in a matched isorecticular series [17], progressive core π -extension promotes exciton dissociation [52], benzobisoxazole-bridged fully conjugated frameworks widen absorption while promoting dissociation [63], fluorenone acceptor cores pair efficient exciton dissociation with a well-defined active center [36], and even 3D COFs, long excluded by their sp^3 nodes, now sustain intrinsic metal-free photocatalysis [102]. Third, heteroatom decoration acts as a local-field lever: fluorination raises surface area, narrows the gap, and improves separation [54], with positional isomerism of fluorine alone moving AQY to 4.09% at 500 nm [143]; chlorine placement creates halogen-adjacent carbon HER sites active in seawater without cocatalyst [61]; and push–pull side groups correlate directly with ESIPT-mediated activity across matched six-membered substituent series [41, 45]. Local polarization engineering in truxenone frameworks [37], dipole modulation of vinylene COFs [98], and nitrogen-position tuning that reverses framework polarity [32] all act on the same underlying variable. The design rule: maximize the net (not nominal) D–A dipole per unit cell, verify with E_b or built-in field measurements, and only then attribute activity gains to the D–A lever [4, 81].

2.3 Linkage protonation and acid engineering

Protonation is the cheapest band-structure intervention available: exposing imine or azine lone pairs to Brønsted acids red-shifts absorption, stabilizes the excited state, and switches on exciton dissociation channels. The landmark demonstration lifted D–A imine COFs to $20.7 \text{ mmol g}^{-1} \text{ h}^{-1}$ upon linkage protonation [114]; HCl-protonated COF-920 subsequently reached $364 \text{ mmol g}^{-1} \text{ h}^{-1}$ under UV–visible light [17, 18]. The lever composes with others: simultaneous protonation and conjugation enhancement multiplies to an 88.4-fold gain over the parent framework [38], surface protonation plus interfacial engineering yields $18.23 \text{ mmol g}^{-1} \text{ h}^{-1}$ without any cocatalyst [20], and protonation raises conductivity and accelerates H_2 generation at Ru sites in vinylene frameworks [123]. The boundary condition is stability: protonation equilibria drift under operation, so durability claims require post-reaction structural verification (Section 7).

2.4 Defect and vacancy engineering

Beyond backbone substitution, deliberately installed defects create charge funnels: amide-functionalized defective CTFs reach $17.6 \text{ mmol g}^{-1} \text{ h}^{-1}$, a 6.3-fold gain over the pristine framework, while preserving structural integrity [29], and missing-linker defects in porphyrin D–A COFs improve hydrogen evolution alongside oxygen reduction [117]. Crystallinity-resolved studies caution that defect and order effects are

entangled: molecular engineering of tunable-crystallinity series shows activity tracking order parameters rather than nominal composition [108], which is why Section 4 treats crystallinity control as a first-class synthesis variable rather than a nuisance.

3 Mechanistic Characterization: Testing the Causal Chain

A design chain is only as credible as the measurements that can falsify each link. This section reviews the toolbox in the order the chain itself proposes – factor isolation first, then carrier dynamics, then atomic-scale and operando interface probes – and adopts calibrated language throughout: spectra *show* observables, global analysis *assigns* components under a model, and matched activity correlations *are consistent with* rather than prove causality [12].

3.1 Factor deconvolution before spectroscopy

Because COF photocatalysts co-vary crystallinity, porosity, absorption, wettability, and charge extraction, single-material reports rarely identify which factor moved. The structure–property–activity protocol of Ghosh and co-workers – quasi-independent variation of factors across the S-COF/FS-COF benchmark family – identified crystallinity and charge extraction as the prime HER determinants and remains the template for benchmarking claims [26, 101]. Substituent series on fixed backbones extend the same logic: six-membered substituent sets isolate electronic effects from framework geometry [45], and matched isorecticular linker series separate linker length from core identity [17]. Reviews of the crystallinity–stability dilemma emphasize that order parameters must be reported alongside rates for any comparison to be meaningful [85, 108].

3.2 Carrier dynamics: from femtoseconds to seconds

Time-resolved spectroscopy provides the quantitative observables of the middle links. Femtosecond transient absorption (fs-TA) resolves exciton formation, dissociation, and trapping; in S-scheme heterojunctions it directly tracks interfacial electron flow, as shown for ZnSe quantum dot/COF [137] and FS-COF/WO₃ hybrids [133]. Time-resolved photoluminescence, surface photovoltage, and wavelength-dependent photocurrent action spectra together assigned the enhanced activity of g-C₃N₄/Ni-CAT junctions to transfer of high-energy-level electrons rather than to absorption gains [13]. Nanoscale COF colloids show molecule-like excitonic behavior in photoluminescence and transient absorption, tying carrier physics to particle dimension [136], and lifetime engineering through D–A pairing is read out as dramatically prolonged carrier lifetimes accompanying the 82.6% AQE benchmark [51]. The historical caution from oxide photocatalysis applies unchanged: lifetime changes alone are not productive charge transfer unless they correlate with quantum yield under matched intensity [62, 94].

3.3 Atomic-scale and operando interface probes

Where the chain meets the cocatalyst, aberration-corrected HAADF-STEM and X-ray absorption fine structure establish speciation: atomically dispersed Pt on TpPa-1 was shown to bind as a six-coordinated C₃N–Pt–Cl₂ species, with DFT connecting that geometry to the activity gain [14]. Soft X-ray absorption identified halogen-adjacent carbons as the active H* binding sites in Cl-decorated frameworks [61], and in situ XPS traced the S-scheme electron pathway in fully conjugated COF/C₃N₄ junctions [57]. Mechanistic assignment of the Volmer–Tafel route with a Tafel rate-determining step on Ru single-atom/nanocluster hybrids shows how kinetic analysis complements spectroscopy [84]. FT-IR and ¹H NMR located hydrogen-bonded nanoreactor encapsulation inside COF pores [75], while ESPT-induced photoisomerization of keto-enamine linkages – invisible to steady-state diffraction – was caught by excited-state spectroscopy, reshaping how activity differences within the Tp family should be interpreted [105].

3.4 What counts as mechanistic evidence

Three practices from the strongest studies deserve adoption as field norms. First, matched controls: physical blends and amorphous analogues distinguish framework effects from composition effects [66, 110]. Second, multi-probe convergence: charge-separation claims supported jointly by fs-TA kinetics, photoelectrochemical response, and activity correlations [40, 89, 127] are categorically stronger than single-technique assignments. Third, protocol-anchored quantum yields: AQE at stated wavelengths – 6.7% at 420 nm for alkene linkages [69], 2.53% for sp^2c overall splitting [11], 9.39% under piezo-assisted operation [22] – remain the only observables portable across laboratories, a theme Section 5 develops quantitatively [139].

4 Synthesis Methodology and Crystallinity Control

Every claim in Sections 2 and 3 is gated by synthesis: crystallinity, stacking order, and morphology are set in the reactor, and they co-determine absorption, exciton transport, and site accessibility [82, 85]. This section organizes the methodology by decreasing thermodynamic control and increasing practicality.

4.1 Solvothermal standards and crystallization modulation

Most COF photocatalysts are still grown solvothermally – sealed-tube, 100–120 °C, days-long reactions whose solvent pair, catalyst, and water activity control the error-correction equilibrium [3, 106]. The Tp toolbox illustrates the maturity of this control: 1,3,5-triformylphloroglucinol reliably delivers crystalline, stable β -ketoenamine frameworks across many diamines, which is a major reason Tp-COFs dominate the composite literature [92, 97, 112]. For imine frameworks, cocrystal engineering of the monomer with p-toluenesulphonic acid regulates crystal structure formation and enables hydrogen-bond-directed growth [5], while base identity and loading tune triazine- sp^2c condensation [125]. Alkalinity screening in sp^2c photocathode synthesis shows the payoff quantitatively: optimal conditions (110 °C, 30 μ L pyridine, 2 days) yield photocurrents 2.5 $\times$ higher than suboptimally crystallized batches of the same composition [44] – a reminder that inter-laboratory rate disagreements often reduce to crystallinity differences [26, 108].

4.2 Fast, mild, and scalable alternatives

The scale-up path runs through routes that abandon the sealed tube. Mechanochemical grinding delivers β -ketoenamine COFs at room temperature with moderate crystallinity and retained chemical stability, and mechanochemically synthesized Tp-COFs now support systematic structure–property studies in photocatalytic HER [2]. Flow synthesis of TpPa-1 guided by green-chemistry principles modulates structure continuously and points toward manufacturing-compatible production [111]. Microwave heating compresses heterojunction growth to minutes [24, 25, 130], and nanoconfined synthesis at surfactant-stabilized oil–water emulsion interfaces reconciles high reaction rate with morphology control for imine 2D COFs [65]. Post-synthetic and irreversible-bond strategies – Katritzky-type skeletal editing among them – expand the accessible linkage space beyond dynamic covalent chemistry [76]. Room-temperature and template-free routes remain the least developed but most scale-relevant frontier [106].

4.3 Stacking stabilization and morphology engineering

Layer slippage and exfoliation degrade the columnar π -channels that carry charge; stabilizing the stack is therefore a synthesis objective in its own right. PEG-threading of pores locks the ordered layer arrangement and preserves HER activity [140]. Downsizing to colloidal nano-COFs multiplies mass-normalized rates – up to 392 mmol $g^{-1} h^{-1}$ – by shortening exciton diffusion paths and exposing surface sites [136]; nanosheet engineering of β -ketoamine frameworks with pore-confined sub-nanometer Pt enables overall water splitting at 0.23% solar-to-hydrogen efficiency [118]. Morphology also mediates composite quality: in situ growth on oxide scaffolds produces chemically bonded, ultrathin junction layers [73, 90], and coordination-directed

Table 2: Representative single-phase COF photocatalysts. Rates in $\text{mmol g}^{-1} \text{h}^{-1}$ unless noted; n.r. = not reported in the audited record.

System (linkage)	Design lever	Light / donor	Cocat.	Rate	AQY / note
Protonated D–A imine COF [114]	linkage protonation	visible / n.r.	Pt	20.7	landmark lever demo
COF-920 (protonated) [17]	isorecticular + protonation	UV–vis / n.r.	n.r.	364	gas-collection verified
β -Ketene–cyano D–A nanosheets [51]	D–A pair, exfoliation	450 nm / n.r.	Pt	n.r.	AQE 82.6%
Nano-COF colloids [136]	nanoscale downsizing	visible / n.r.	Pt	392	$33.3 \mu\text{mol h}^{-1}$ absolute
DBC-sp ² c-COF [122]	sp ² c Kagome lattice	visible / n.r.	Pt	105.3	80.0 in natural seawater
COF-954 [139]	oligo side chains	visible / n.r.	5 wt% Pt	137.2	191.7 (seawater, UV–vis)
TAPPy-DBTDP-COF [55]	dual acceptors	visible / n.r.	Pt	12.7	top D–A tier
TpPaCl [61]	halogen active sites	visible / n.r.	none	16.3	11.5 in real seawater
TP-COF-F-3 [143]	F positional isomer	500 nm / n.r.	n.r.	n.r.	AQY 4.09%
COF–alkene [69]	linkage isolation	420 nm / n.r.	Pt	n.r.	AQE 6.7%
d-CTF(CONH ₂) [29]	designed defects	visible / n.r.	n.r.	17.6	6.3× vs. CTF-1
Pt@TpBpy-NS [118]	nanosheet + pore-confined Pt	visible / none	sub-nm Pt	n.r.	OWS, STH 0.23%
sp ² c-COF (OWS) [11]	full conjugation	420 nm / none	n.r.	n.r.	AQE 2.53%, OWS

assembly of MOF-on-COF architectures yields intimate interfacial contact [64, 79]. Porphyrin-COF platforms carrying single-atom metals show that site placement can be encoded synthetically rather than post-hoc [9, 39].

The methodological bottom line mirrors the mechanistic one: report the order parameters (PXRD width, surface area, stacking signature) with the same prominence as the rate, because the synthesis variables are the hidden covariates of the entire literature [26, 82].

5 Performance Comparison Under Explicit Protocols

5.1 Why raw rates do not rank materials

Reported HER rates for COF systems span from below $1 \text{ mmol g}^{-1} \text{h}^{-1}$ to nearly $400 \text{ mmol g}^{-1} \text{h}^{-1}$, but the protocol variables – irradiation spectrum and intensity, sacrificial donor, cocatalyst identity and loading, catalyst mass, and dispersion state – move rates by larger factors than most design levers [26, 136, 139]. Mass-normalized rates fall as loading rises; apparent quantum yield (AQY) at a stated wavelength and solar-to-hydrogen (STH) efficiency are the only cross-laboratory portable metrics [11, 59]. Pt-size studies underline the danger of single-variable narratives: across three Tp-series COFs the photodeposited Pt particle size varied with band gap, yet HER rates showed no correlation with that size [100]. Tables 2 and 3 therefore list protocol columns explicitly and mark unreported fields as n.r.; entries are exemplars, not a leaderboard.

Table 3: Representative COF heterojunction and composite systems. Abbreviations: ZIS = ZnIn_2S_4 ; BP = black phosphorus.

System	Junction type	Light / donor	Cocat.	Rate	Note
Pt/S@TpPa-1 [83]	Schottky, low work function	420 nm / n.r.	Pt	n.r.	AQE 80.7%
Py-HOF/COF [23]	organic–organic	visible / n.r.	n.r.	390.7	9.2× vs. COF
NiCo-NC@TP-BD [75]	pore nanoreactor	visible / n.r.	none	79.0	395× vs. bare COF
TiO ₂ NFs@COF [25]	S-scheme	visible / n.r.	n.r.	58.3	microwave growth
Ru(II)PS@Py-COF [127]	molecular complex	visible / n.r.	n.r.	51.3	2.5× vs. COF
Diquat-ETM COF [66]	integrated e [−] mediator	visible / n.r.	n.r.	34.6	beats physical mix
CdS/TFPT-DHHTH-COF [131]	direct Z-scheme	visible / n.r.	none	15.8	75× vs. COF
CdS/TpPa-COF (1D) [21]	S-scheme, directional	visible / n.r.	n.r.	n.r.	TpPa platform
P-COF-1/CTF [110]	2D–2D π – π	visible / n.r.	n.r.	14.1	2.5× vs. CTF
JUC-704@ZIS [128]	cocatalyst-free junction	visible / n.r.	none	9.46	seawater operation
HKUST-1/TpPa-1 [64]	MOF-on-COF, II-scheme	visible / n.r.	n.r.	n.r.	coordination-bonded interface
BP/COF (2D–2D) [113]	metal-free p–n	visible / n.r.	none	n.r.	cocatalyst-free

5.2 The cocatalyst-dependence spectrum

Because Pt cost and scarcity bound any deployment scenario [84], the field is best read as a progression along a cocatalyst-dependence axis. At the noble-metal end, engineering focuses on atom economy: single-atom Pt anchored through defined C_3N –Pt– Cl_2 coordination on TpPa-1 [14], heteroatom-tuned Pt single-atom frameworks [95], sulfur-chelated high-loading single atoms that remain atom-efficient ($11.4 \text{ mmol g}^{-1} \text{ h}^{-1}$ at 12.1 wt%) [7], and low-work-function Pt junctions whose $\Delta G_{\text{H}^+} = -0.13 \text{ eV}$ optimality yields AQE 80.7% at 420 nm [83]. Earth-abundant single atoms follow: Co on COFs [103], Fe–Co dual atoms on COF/ C_3N_4 [74], zinc single atoms bridged by copper clusters [99], spinel cocatalysts with engineered orbital interactions [126], and DFT screens that rationalize single-atom–support pairing [72, 121]. Molecular and cluster catalysts occupy the middle ground: cyclometalated Pt in vinylene frameworks [46], azide-anchored cobaloximes on COF-42 that outlast their physisorbed analogues [27], enantioselectively immobilized chiral Co complexes ($5.70 \text{ mmol g}^{-1} \text{ h}^{-1}$ noble-metal-free) [42], Cu(II) mediators on chiral β -ketoenamine hosts [104], and Ru sites on triazine sp^2c platforms extending to electrocatalytic HER [123, 135, 138]. The cocatalyst-free end is growing fastest: halogen-site TpPaCl [61], protonated CdS/COF architectures [20], benzothiazole metal-COFs [134], JUC-70x@ZIS junctions in seawater [128, 129], black-phosphorus p–n junctions [113], and C_{60} @TpPa charge-transfer dyads [124].

5.3 Stability reporting

Durability claims currently mix hour-scale runs, cycle counts, and post-mortem characterization. The stronger reports pair retention percentages with structural verification: 12-hour photocurrent stability

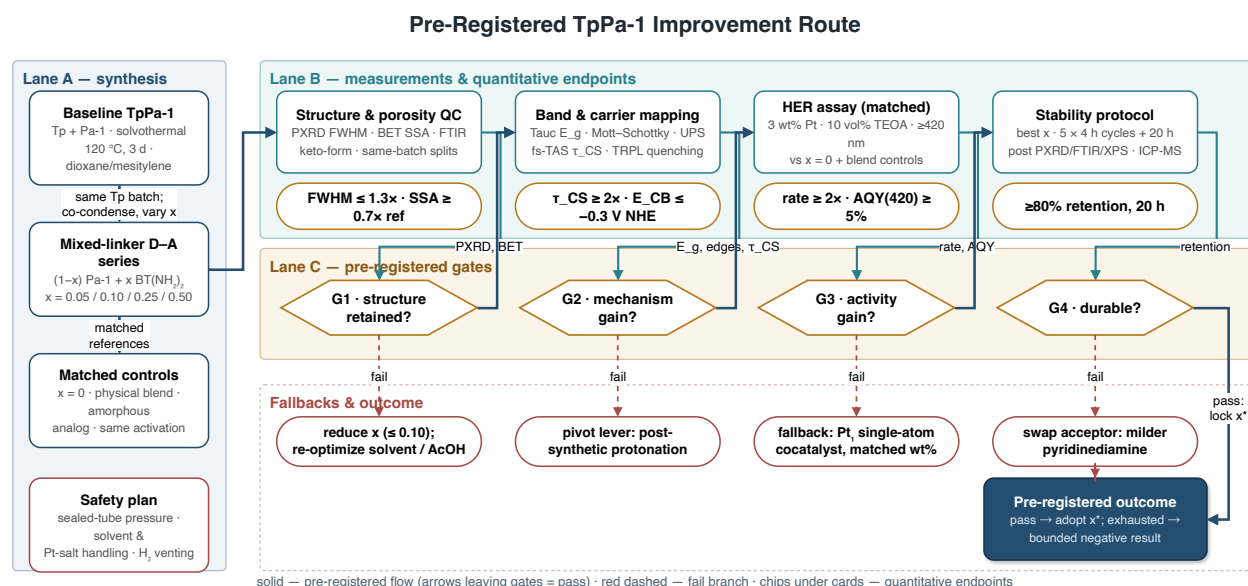


Figure 2: Pre-registered TpPa-1 improvement route. Lane A fixes synthesis and matched controls; Lane B specifies quantitative measurement endpoints; Lane C gates progression (G1–G4). Red dashed branches are pre-committed fallbacks; the outcome is adopted only if all gates pass, otherwise the result is reported as a bounded negative.

with retained crystallinity [58], sustained performance of integrated-interface frameworks [40], S-scheme junctions surviving repeated cycling [15, 16, 57, 88, 132, 141], and functional-group- resolved junction dynamics [31, 109]. Chirality-induced spin selectivity [107], piezo-assisted charge steering [22], and interfacial covalent bonding [50, 79, 119] each claim durability side benefits that still await standardized, long-duration verification – the protocol gap Section 7 addresses [19, 35, 93].

6 A Pre-Registered Improvement Route for TpPa-1

The design rules of Sections 2–5 are only useful if they generate falsifiable predictions. This section formulates one: a mixed-linker donor–acceptor doping route for TpPa-1, chosen because the baseline is the field’s best-characterized β -ketoenamine platform [92, 97, 112] – stable in boiling water and strong acid, synthesizable solvothermally, mechanochemically, and in flow [2, 111] – yet its interrupted in-plane conjugation and modest exciton dissociation leave clear headroom that composites routinely exploit [16, 21, 64, 83, 124, 141].

6.1 Hypothesis and route

The causal chain predicts that installing a fraction x of an electron-accepting diamine – benzothiadiazole-diamine, BT(NH₂)₂ – into the Pa-1 position creates local D–A dipoles that lower the exciton binding energy and raise charge-separation yield, without abandoning the tautomer-locked stability of the Tp node (Figure 2, Lane A). The acceptor choice has direct precedent: benzothiadiazole-dianiline frameworks are established HER platforms whose activity responds sharply to local electronic modulation [10, 55]. Statistical copolymerization at $x \in \{0.05, 0.10, 0.25, 0.50\}$ under fixed solvothermal conditions (120 °C, 3 d, dioxane/mesitylene) holds the linkage chemistry constant while sweeping the dipole density – the quasi-independent factor-variation design that benchmark studies established [17, 26, 45]. Matched controls are pre-committed: the $x = 0$ baseline, a physical blend, and an amorphous analogue with identical composition and activation [66, 110].

6.2 Quantitative gates

Lane B fixes the endpoints before synthesis begins, and Lane C converts them into pass/fail gates. G1 (structure retained): PXRD line width within $1.3\times$ of baseline and BET surface area at least $0.7\times$ baseline, so that crystallinity – the dominant hidden covariate [44, 108] – cannot masquerade as an electronic effect. G2 (mechanism gain): charge-separation lifetime $\tau_{CS} \geq 2\times$ baseline by fs-TA with TRPL-quenching corroboration, and conduction-band edge at or below -0.3 V vs. NHE by Tauc/Mott–Schottky/UPS so the HER driving force is preserved [12, 13, 81]. G3 (activity gain): HER rate at least $2\times$ baseline and AQY(420 nm) $\geq 5\%$ under a matched assay (3 wt% Pt, 10 vol% TEOA, ≥ 420 nm) benchmarked against the $x = 0$ and blend controls [69, 100]. G4 (durability): $\geq 80\%$ rate retention over 20 h in 5×4 h cycles with post-run PXRD/FTIR/XPS/ICP-MS verification, the stability standard the composite literature is converging toward [40, 58].

6.3 Fallbacks and interpretation

Each gate failure triggers a pre-committed pivot rather than post-hoc tuning (Figure 2, bottom): G1 failure reduces x and re-optimizes solvent acidity; G2 failure pivots the lever to post-synthetic protonation, which acts on the same link of the chain with independent precedent [20, 38, 114]; G3 failure swaps in an atom-economic single-atom cocatalyst at matched loading [7, 14]; G4 failure substitutes the milder pyridine-diamine acceptor. If all gates pass, the adopted x^* becomes the new baseline; if the fallback tree is exhausted, the result is reported as a bounded negative – the D–A dipole density is insufficient at preserved crystallinity for this family – which is itself a publishable constraint on the causal chain [26]. Safety is pre-registered alongside science: sealed-tube pressure, solvent handling, Pt-salt disposal, and H_2 venting follow the protocols embedded in the route plan.

7 Challenges on the Path to Scalable Solar Fuels

Following the style of the strongest outlooks, we phrase each challenge as a decision question with a measurement and a failure criterion rather than as a wish list [19, 93].

Can the field converge on portable metrics? The comparison problem of Section 5 is structural: rates without wavelength-resolved AQY, donor identity, and normalization basis cannot be aggregated [100, 139]. The decision question is whether journals and reviews require AQY(λ) and STH – with light-intensity linearity checks inherited from oxide photocatalysis [94] – as submission defaults. Failure looks like another decade of leaderboards built on incommensurable $\text{mmol g}^{-1} \text{h}^{-1}$ values [1, 28].

Does activity survive the donor? Nearly all Table 2 entries consume sacrificial electron donors. Overall water splitting on COF platforms exists – sp^2c frameworks at AQE 2.53% [11], pore-confined Pt nanosheets at STH 0.23% [118], bifunctional Pt/COF systems evolving stoichiometric H_2/O_2 [130], and computationally paired HER/OER COF couples [96] – but efficiencies trail donor-assisted numbers by two orders of magnitude. The measurable milestone is STH $\geq 1\%$ on an all-organic or COF-dominant system with verified stoichiometry; persistent reliance on TEOA-class donors within techno-economic models is the failure mode [6, 43].

Is long-duration stability real? Hour-scale retention with post-mortem PXRD/XPS is becoming standard [40, 58], but solar-fuel service means thousands of hours under flow, oxygen crossover, and protonation drift (Section 2.3). The decision experiment is a 1000-hour continuous run with periodic structural audit – the COF analogue of module aging tests; crystallinity–stability trade-offs [82, 85] and linkage hydrolysis under operando acidity are the leading failure hypotheses.

Can synthesis leave the sealed tube? Gram-scale, days-long solvothermal batches cannot supply reactor arrays. Flow synthesis of TpPa-1 [111], mechanochemical batches [2], microwave composite growth [24], and emulsion-interface morphology control [65] define the candidate manufacturing set; the milestone is kilogram-scale production retaining Section 4 order parameters at $<10\%$ penalty [49, 106].

Can noble metals be engineered out? The Pt-dependence spectrum of Section 5 is trending in the right direction – single-atom economy [14, 95], earth- abundant single atoms [74, 103, 121], molecular Co catalysts [27, 42], and genuinely cocatalyst-free platforms [61, 113, 128, 134] – but matched-protocol parity with 3 wt% Pt baselines remains the unambiguous target [84].

Do photons reach the framework in a reactor? Photoelectrode and panel formats change the photon economics that powder assays hide: COF photocathodes sustain tens of $\mu\text{A cm}^{-2}$ [44, 58], oxide-scaffolded junctions manage light and charge jointly [35, 73], and PEC-oriented reviews now treat 2D COFs as absorber layers in their own right [30, 77]. Reactor-level milestones – areal STH under natural irradiance, seawater tolerance [122, 128, 139] – should replace mass-normalized records as the field’s headline numbers [72, 142].

8 Conclusion

Organizing a decade of COF photocatalysis along a single causal chain – linkage $\rightarrow$ conjugation $\rightarrow$ band structure $\rightarrow$ exciton and charge separation $\rightarrow$ interface $\rightarrow$ HER – turns a sprawling literature into a set of testable design rules. Linkage chemistry no longer forces a stability–conjugation trade-off: vinylene and sp^2 -carbon frameworks deliver both, while β -ketoenamine and imine platforms remain the workhorses for systematic study [11, 47, 112]. Donor–acceptor dipole engineering, verified by exciton-binding measurements rather than nominal composition, and linkage protonation are the most portable band-structure levers [51, 81, 114]. Mechanistic claims increasingly rest on multi-probe convergence with matched controls [13, 26], and the strongest performance reports anchor rates to $\text{AQY}(\lambda)$ with explicit protocol variables [83, 136]. What stands between laboratory quantum efficiency and solar fuels is no longer molecular designability but verification infrastructure: portable metrics, donor-free operation, thousand-hour stability, scalable synthesis, and reactor-level photon management [93, 106]. The pre-registered TpPa-1 route of Section 6 is offered as a template for how the field can convert its accumulated design rules into gated, falsifiable experiments – and how negative results, bounded by matched controls, can constrain the causal chain as productively as records [69, 100].

References

- [1] Awais Ali Aslam, Muhammad Shahid Nazir, Sadaf ul Hassan, Majid Niaz Akhtar, Murid Hussain, Sania Amjad, Alia Farooq, and Aqeel Ahmad Khan. A review on recent developments towards hydrogen evolution reaction for covalent organic frameworks. In *Renewable and Sustainable Energy Reviews*, 2026. doi: 10.1016/j.rser.2025.116695. URL <https://doi.org/10.1016/j.rser.2025.116695>.
- [2] Kiran Asokan and Sukumaran Santhosh Babu. Structure–property relation of mechanochemically synthesized β -ketoenamine-linked COFs in photocatalytic hydrogen evolution. *Chemistry – An Asian Journal*, 2026. doi: 10.1002/asia.70886. URL <https://doi.org/10.1002/asia.70886>.
- [3] Ahmad Reza Bagheri and Nahal Aramesh. Towards the room-temperature synthesis of covalent organic frameworks: a mini-review. *Journal of Materials Science*, 2020. doi: 10.1007/s10853-020-05308-9. URL <https://doi.org/10.1007/s10853-020-05308-9>.
- [4] Yun-Yun Bao, Qiao qiao Jiang, Xun Wang, Zhen-Wen Zhang, Bin Lin, Cheng-Rong Zhang, Ru-Ping Liang, Feng tao Yu, and Jian-Ding Qiu. Excitonic engineering in low-symmetry covalent organic frameworks through π -n lewis pair integration for enhanced photocatalytic performance. *Journal of Colloid and Interface Science*, 2025. doi: 10.1016/j.jcis.2025.139822. URL <https://doi.org/10.1016/j.jcis.2025.139822>.
- [5] Hongguang Chen, Daming Feng, Fengxia Wei, Fang Guo, and Anthony K. Cheetham. Hydrogen-bond-regulated mechanochemical synthesis of covalent organic frameworks: cocrystal precursor strategy for confined assembly. In *Angewandte Chemie*, 2024. doi: 10.1002/anie.202415454. URL <https://www.semanticscholar.org/paper/9a0e79297c64e7baef10c71467b18278477f6357>.
- [6] Liangji Chen and Jier Huang. Covalent organic framework photocatalysts for water oxidation and overall artificial photosynthesis. In *ACS Nano*, 2026. doi: 10.1021/acsnano.5c22772. URL <https://www.semanticscholar.org/paper/7f6ad292a6a2f74db3dab33985df8a92ea8b5c5c>.
- [7] Liangjun Chen, Guinan Chen, Chengtao Gong, Yifei Zhang, Zhihao Xing, Jiahao Li, Guodong Xu, Gao Li, and Yongwu Peng. Low-valence platinum single atoms in sulfur-containing covalent organic frameworks for photocatalytic hydrogen evolution. In *Nature Communications*, 2024. doi: 10.1038/s41467-024-54959-8. URL <https://doi.org/10.1038/s41467-024-54959-8>.
- [8] Rufan Chen, Ji-Long Shi, Yuan Ma, Guiqing Lin, Xianjun Lang, and Cheng Wang. Designed synthesis of a 2D porphyrin-based sp^2 carbon-conjugated covalent organic framework for heterogeneous photocatalysis. In *Angewandte Chemie International Edition*, 2019. doi: 10.1002/anie.201902543. URL <https://doi.org/10.1002/anie.201902543>.
- [9] Rufan Chen, Yang Wang, Yuan Ma, Arindam Mal, Xiaoya Gao, Lei Gao, Lijie Qiao, Xu-Bing Li, Li-Zhu Wu, and Cheng Wang. Rational design of isostructural 2D porphyrin-based covalent organic frameworks for tunable photocatalytic hydrogen evolution. In *Nature Communications*, 2021. doi: 10.1038/s41467-021-21527-3. URL <https://doi.org/10.1038/s41467-021-21527-3>.
- [10] Yuze Chen, Liyi Yao, Xiaowei Song, Tianjun Zhang, and Zhiqiang Liang. Modulating coordination environment of single nickel atom in covalent organic framework to enhance photocatalytic hydrogen evolution reaction. In *Inorganic Chemistry*, 2025. doi: 10.1021/acs.inorgchem.4c05494. URL <https://www.semanticscholar.org/paper/3f5bd5bb56bd07c1fda6aaa9c68b29176860d49c>.

- [11] Jun Cheng, Yuting Wu, Wei Zhang, Jie Zhang, Lei Wang, Lei Wang, Meng Zhou, Fengtao Fan, Xiaojun Wu, and Hangxun Xu. Fully conjugated 2D sp² carbon-linked covalent organic frameworks for photocatalytic overall water splitting. In *Advanced Materials*, 2023. doi: 10.1002/adma.202305313. URL <https://doi.org/10.1002/adma.202305313>.
- [12] Jinhyuk Choi, Wooteak Jung, Soranyel González-Carrero, James R. Durrant, Hyojung Cha, and Taiho Park. Understanding charge carrier dynamics in organic photocatalysts for hydrogen evolution. In *Energy & Environmental Science*, 2024. doi: 10.1039/d4ee01808g. URL <https://doi.org/10.1039/d4ee01808g>.
- [13] Xiaoyu Chu, Bing-Bing Luan, Ao-Xiang Huang, Yongkuo Zhao, Hongxia Guo, Ning Yang, Haojian Cheng, Guiling Zhang, and Fengming Zhang. Controlled synthesis of 2D–2D conductive metal–organic framework/g-c 3 n 4 heterojunctions for efficient photocatalytic hydrogen evolution. *Dalton Transactions*, 2024. doi: 10.1039/d3dt03894g. URL <https://doi.org/10.1039/d3dt03894g>.
- [14] Pengyu Dong, Yan Wang, Aicaijun Zhang, Ting Cheng, Xinguo Xi, and Jinlong Zhang. Platinum single atoms anchored on a covalent organic framework: Boosting active sites for photocatalytic hydrogen evolution. In *ACS Catalysis*, 2021. doi: 10.1021/acscatal.1c03441. URL <https://doi.org/10.1021/acscatal.1c03441>.
- [15] Pengyu Dong, Aicaijun Zhang, Ting Cheng, Jinkang Pan, Jun Tae Song, Lei Zhang, Rongfeng Guan, Xinguo Xi, and Jinlong Zhang. 2D/2D S-scheme heterojunction with a covalent organic framework and g-C₃N₄ nanosheets for highly efficient photocatalytic H₂ evolution. *CHINESE JOURNAL OF CATALYSIS (CHINESE VERSION)*, 2022. doi: 10.1016/s1872-2067(22)64094-4. URL [https://doi.org/10.1016/s1872-2067\(22\)64094-4](https://doi.org/10.1016/s1872-2067(22)64094-4).
- [16] Pengyu Dong, Ting Cheng, Jinlong Zhang, Jinlong Zhang, Jinhui Jiang, Lei Zhang, Xinguo Xi, Jinlong Zhang, and Jinlong Zhang. Fabrication of an organic/inorganic hybrid TpPa-1-COF/ZnIn₂S₄ s-scheme heterojunction for boosted photocatalytic hydrogen production. In *ACS Applied Energy Materials*, 2023. doi: 10.1021/acsaem.2c03806. URL <https://doi.org/10.1021/acsaem.2c03806>.
- [17] Wenbo Dong, Xingjiang Yu, Zhiying Qin, Yushuang Chen, Shijie Ren, and Longyu Li. Synergistic enhancement of photocatalytic hydrogen evolution in covalent organic frameworks via isorecticular design, isomerism, and protonation. In *Angewandte Chemie International Edition*, 2025. doi: 10.1002/anie.202511200. URL <https://doi.org/10.1002/anie.202511200>.
- [18] Wenbo Dong, Xingjiang Yu, Zhiying Qin, Yushuang Chen, Shijie Ren, and Longyu Li. Synergistic enhancement of photocatalytic hydrogen evolution in covalent organic frameworks via isorecticular design, isomerism, and protonation. In *Angewandte Chemie*, 2025. doi: 10.1002/ange.202511200. URL <https://doi.org/10.1002/ange.202511200>.
- [19] Jared G. Doremus, Bertha Lotsi, Aadarsh Sharma, and Psaras L. McGrier. Photocatalytic applications of covalent organic frameworks: synthesis, characterization, and utility. In *Nanoscale*, 2024. doi: 10.1039/d4nr03204g. URL <https://doi.org/10.1039/d4nr03204g>.
- [20] Fang Duan, Jialiang Sheng, Songhu Shi, Yujie Li, Wenhao Liu, Shuanglong Lu, Han Zhu, Mingliang Du, Xin Chen, and Jun Wang. Protonated Z-scheme CdS-covalent organic framework heterojunction with highly efficient photocatalytic hydrogen evolution. *Journal of Colloid and Interface Science*, 2024. doi: 10.1016/j.jcis.2024.07.051. URL <https://www.semanticscholar.org/paper/9be27a4eb9032d76a202682f553a088da0ab55e4>.

- [21] Minwei Fan, An Qian, Xin Han, Jihui Yu, Chong Zhang, Lei Ye, Xin Pu, Moldyr Dyusebaeva, Gulzat Berganayeva, Qiang Yang, and Jichang Liu. One-dimensional CdS/TpPa-COF s-scheme heterojunctions enabling directional charge transport for highly efficient photocatalytic hydrogen evolution. In *Advanced Functional Materials*, 2026. doi: 10.1002/adfm.75647. URL <https://doi.org/10.1002/adfm.75647>.
- [22] Jinglin Gao, Jianyu Xiao, Shijia Luo, Xuzheng Ji, Congcong Yin, Yuping Wu, Xin Zhao, and Yong Wang. Piezoceramics boosting photocatalytic hydrogen evolution of covalent organic frameworks. In *Advanced Science*, 2025. doi: 10.1002/advs.202508062. URL <https://doi.org/10.1002/advs.202508062>.
- [23] Rui Gao, Rongchen Shen, Can Huang, Kaihui Huang, Guijie Liang, Peng Zhang, and Xin Li. 2D/2D hydrogen-bonded organic frameworks/covalent organic frameworks s-scheme heterojunctions for photocatalytic hydrogen evolution. In *Angewandte Chemie*, 2024. doi: 10.1002/ange.202414229. URL <https://doi.org/10.1002/ange.202414229>.
- [24] Tengyuan Gao, Xiufan Liu, Qingshan Feng, Xinhe Wu, Juan Wang, and Guohong Wang. Microwave assisted rapid synthesis of tio2 nfs@cof s-scheme heterojunction photocatalyst for highly efficient photocatalytic hydrogen evolution, 2025. URL <https://doi.org/10.2139/ssrn.5133143>.
- [25] Tengyuan Gao, Xiufan Liu, Qingshan Feng, Xinhe Wu, Juan Wang, Juan Wang, and Guohong Wang. Microwave assisted rapid synthesis of TiO₂ nfs@COF S-scheme heterojunction photocatalyst for highly efficient photocatalytic hydrogen evolution. *Journal of Colloid and Interface Science*, 2025. doi: 10.1016/j.jcis.2025.138075. URL <https://doi.org/10.1016/j.jcis.2025.138075>.
- [26] Samrat Ghosh, Akinobu Nakada, Maximilian A. Springer, Takahiro Kawaguchi, Katsuaki Suzuki, Hironori Kaji, Igor A. Baburin, Agnieszka Kuc, Thomas Heine, Hajime Suzuki, Ryu Abe, and Shu Seki. Identification of prime factors to maximize the photocatalytic hydrogen evolution of covalent organic frameworks. *Journal of the American Chemical Society*, 2020. doi: 10.1021/jacs.0c02633. URL <https://doi.org/10.1021/jacs.0c02633>.
- [27] K. Gottschling, Gökcen Savasci, Hugo A. Vignolo-González, S. Schmidt, Philipp Mauker, Tanmay Banerjee, Petra Rovó, C. Ochsenfeld, and B. Lotsch. Rational design of covalent cobaloxime-covalent organic framework hybrids for enhanced photocatalytic hydrogen evolution. *Journal of the American Chemical Society*, 2020. doi: 10.1021/jacs.0c02155. URL <https://www.semanticscholar.org/paper/0f8cd91b1cb7918ba39033e709d20154743d413d>.
- [28] Chang-Cheng Gu, Fenghua Xu, Wei-Kang Zhu, Run-Juan Wu, Lu Deng, Jun Zou, Baicheng Weng, and Rilong Zhu. Recent advances on covalent organic frameworks (COFs) as photocatalysts: different strategies for enhancing hydrogen generation. In *Chemical Communications*, 2023. doi: 10.1039/d3cc01970e. URL <https://doi.org/10.1039/d3cc01970e>.
- [29] Junyi Gu, Lu Liu, Weijia An, Qiao Zhao, Yunqi Li, Shanfu Lu, Sheng Zhang, Xiao Wang, Wenquan Cui, and Baiyan Li. Single-site defect engineering in covalent organic frameworks enables extended directional charge transfer pathway for efficient photocatalytic hydrogen evolution. In *Advanced Functional Materials*, 2026. doi: 10.1002/adfm.77033. URL <https://www.semanticscholar.org/paper/031d9e90f10082db15e7085aef1bed908517b42f>.
- [30] R. Guntermann, Shuo Sun, Jonathan Kampmann, and D. Medina. Recent advances in photoelectrochemical water splitting enabled by covalent organic frameworks. In *Small Structures*, 2026. doi: 10.1002/ssstr.202500844. URL <https://www.semanticscholar.org/paper/6c39bd559a32ec5bd2c1a3b76555fae0523883f7>.

- [31] Hongyan Guo, Shiyong Wang, Xin Chen, Jingfang Kou, Guoqiang He, Zhengping Dong, and Yong Yan. Engineering a covalent organic framework-based type-II heterojunction for enhanced photocatalytic H₂O₂ synthesis. In *Nature Synthesis*, 2025. doi: 10.1038/s44160-025-00880-x. URL <https://www.semanticscholar.org/paper/68184152e2527861a473b2f34ee42c69060b49c9>.
- [32] Frederik Haase, Tanmay Banerjee, Gökçen Savaşçı, Christian Ochsenfeld, and Bettina V. Lotsch. Structure–property–activity relationships in a pyridine containing azine-linked covalent organic framework for photocatalytic hydrogen evolution. In *Faraday Discussions*, 2017. doi: 10.1039/c7fd00051k. URL <https://doi.org/10.1039/c7fd00051k>.
- [33] Huimin Hao, Fulin Zhang, Xiaoyun Dong, and Xianjun Lang. 2D sp² carbon-conjugated triazine covalent organic framework photocatalysis for blue light-induced selective oxidation of sulfides with O₂. In *Applied Catalysis B: Environmental*, 2021. doi: 10.1016/j.apcatb.2021.120691. URL <https://www.semanticscholar.org/paper/8e0ad591a14637f8009b005aea36d2630936b372>.
- [34] Huimin Hao, Fulin Zhang, Xiaoxiao Wang, Fengwei Huang, Shengli Chen, and Xianjun Lang. A 2D sp² carbon conjugated pyrene covalent organic framework with tempo for selective photocatalysis. In *Advanced Functional Materials*, 2024. doi: 10.1002/adfm.202419735. URL <https://www.semanticscholar.org/paper/587e003c51bedc1d731334b002e4ce80bd21767a>.
- [35] K. Hao, A. Jalil, and R. Jusoh. Covalent organic frameworks and TiO₂ nanotube array-based framework photocatalysts for solar hydrogen production: a mini review. *Journal of nanoparticle research*, 2025. doi: 10.1007/s11051-025-06498-6. URL <https://www.semanticscholar.org/paper/77954f82bcbbfe18b5fba23b19ea51fa0c337aaf>.
- [36] Lei Hao, Rongchen Shen, Can Huang, Zi-Zhan Liang, Neng Li, Peng Zhang, Xiuzhi Li, Xiuzhi Li, Chaochao Qin, Xin Li, and Xin Li. Fluorenone-based covalent organic frameworks with efficient exciton dissociation and well-defined active center for remarkable photocatalytic hydrogen evolution. In *Applied Catalysis B: Environmental*, 2023. doi: 10.1016/j.apcatb.2023.122581. URL <https://doi.org/10.1016/j.apcatb.2023.122581>.
- [37] Lei Hao, Rongchen Shen, Chaochao Qin, Neng Li, Haobin Hu, Guijie Liang, and Xin Li. Regulating local polarization in truxenone-based covalent organic frameworks for boosting photocatalytic hydrogen evolution. In *Science China Materials*, 2024. doi: 10.1007/s40843-023-2747-6. URL <https://doi.org/10.1007/s40843-023-2747-6>.
- [38] Huan He, Rongchen Shen, Yuhao Yan, Dejun Chen, Zhixiong Liu, Lei Hao, Xin Zhang, Peng Zhang, and Xin Li. Double enhancement of protonation and conjugation in donor–imine–donor covalent organic frameworks for photocatalytic hydrogen evolution. In *Chemical Science*, 2024. doi: 10.1039/d4sc07028c. URL <https://doi.org/10.1039/d4sc07028c>.
- [39] Ting He, Zhanfeng Zhao, Ruoyang Liu, Xinyan Liu, Bing Ni, Yanping Wei, Yinglong Wu, Wei Yuan, Hong-Jie Peng, Zhongyi Jiang, and Yanli Zhao. Porphyrin-based covalent organic frameworks anchoring Au single atoms for photocatalytic nitrogen fixation. *Journal of the American Chemical Society*, 2023. doi: 10.1021/jacs.2c10233. URL <https://doi.org/10.1021/jacs.2c10233>.
- [40] Ting He, Wenlong Zhen, Yongzhi Chen, Yuanyuan Guo, Zhuoer Li, Ning Huang, Zhongping Li, Ruoyang Liu, Yuan Liu, X. Lian, Can Xue, T. Sum, Wei Chen, and Donglin Jiang. Integrated interfacial design of covalent organic framework photocatalysts to promote hydrogen evolution from water. In *Nature Communications*, 2023. doi: 10.1038/s41467-023-35999-y. URL <https://www.semanticscholar.org/paper/6ed415419faa23edfcb6997a8f36747f94484c5>.

- [41] Xingye Huang and Jia Guo. Regulating the photoisomerization of covalent organic framework for enhanced photocatalytic hydrogen evolution†. *Chinese journal of chemistry*, 2024. doi: 10.1002/cjoc.202400437. URL <https://www.semanticscholar.org/paper/695f4e69db8a58b53617524a360c20db5e15a5c3>.
- [42] Xingye Huang, Wanting Xie, Tiantian Xu, Weijun Weng, Ting Zhou, and Jia Guo. Enantioselective immobilization of nonprecious metal complexes on chiral covalent organic frameworks for improved single-site photocatalytic hydrogen evolution. In *Angewandte Chemie*, 2025. doi: 10.1002/anie.202509095. URL <https://www.semanticscholar.org/paper/e29e3b51d4067f3ba6c927854023ada9c15e7419>.
- [43] Baodong Jia, Long-Sheng Cai, Zhi-Hao Zhao, Xin Jin, Ya Wang, Hong Dong, and Feng-Ming Zhang. Rational design pyridine-based covalent organic framework S-scheme heterojunction for enhanced photocatalytic overall water splitting. In *Applied Surface Science*, 2027. doi: 10.1016/j.apsusc.2026.168160. URL <https://doi.org/10.1016/j.apsusc.2026.168160>.
- [44] Ziyang Jia, Yang Lu, Wenyan Li, Yiyi Fan, Yawen Tang, Lei Cheng, and Hanjun Sun. Copper-surface-mediated crystallization engineering of sp² carbon-conjugated covalent organic framework for photoelectrochemical hydrogen evolution. In *Solar RRL*, 2025. doi: 10.1002/solr.202500636. URL <https://www.semanticscholar.org/paper/f65cb65ce0606398277ba2ef07587a31b8b30825>.
- [45] Shaodong Jiang, Hongyun Niu, Qing Sun, Rusong Zhao, Na Li, and Yaqi Cai. Significant improvement of photocatalytic hydrogen evolution performance in covalent organic frameworks: substituent fine-tuning. *Journal of Materials Chemistry A*, 2024. doi: 10.1039/d4ta00478g. URL <https://doi.org/10.1039/d4ta00478g>.
- [46] Xiang-Ning Jiao, Yuan-Zhe Cheng, Yuan-Yuan Duan, Xiaotian Bao, Dong-Hui Yang, Xinfeng Liu, Bao-Hang Han, and Xuesong Ding. Cyclometalated platinum in a vinylene-linked covalent organic framework for photocatalytic hydrogen evolution. *Chemical Engineering Journal*, 2026. doi: 10.1016/j.cej.2026.179128. URL <https://doi.org/10.1016/j.cej.2026.179128>.
- [47] Enquan Jin, Zhian Lan, Qihong Jiang, Keyu Geng, Guosheng Li, Xinchun Wang, and Donglin Jiang. 2D sp² carbon-conjugated covalent organic frameworks for photocatalytic hydrogen production from water. In *Chem*, 2019. doi: 10.1016/j.chempr.2019.04.015. URL <https://doi.org/10.1016/j.chempr.2019.04.015>.
- [48] Yucheng Jin, Heyuan Liu, Qixin Zhou, Xiaoning Zhan, Boce Cui, Hailong Wang, Dongdong Qi, Zhiying Chen, Baoqiu Yu, Xiaonan Pang, Bin Liu, and Jianzhuang Jiang. Abnormal multistep charge transfer in a 2D single-crystalline covalent organic framework photocatalyst. *Journal of the American Chemical Society*, 2026. doi: 10.1021/jacs.6c08611. URL <https://www.semanticscholar.org/paper/7322588a61be625b7e8ec1834f2462b6cf6575c1>.
- [49] Islam E. Khalil, Prasenjit Das, and Arne Thomas. Two-dimensional covalent organic frameworks: Structural insights across different length scales and their impact on photocatalytic efficiency. In *Accounts of Chemical Research*, 2024. doi: 10.1021/acs.accounts.4c00491. URL <https://doi.org/10.1021/acs.accounts.4c00491>.
- [50] Bingzhu Li, Xiaohua Ma, Minjun Lei, Xiaoli Ma, Youji Li, and Zhiliang Jin. Modifying covalent organic frameworks with mos₂ form Schottky heterojunctions for efficient photocatalytic hydrogen production. *Journal of Power Sources*, 2025. doi: 10.1016/j.jpowsour.2025.238364. URL <https://www.semanticscholar.org/paper/5a3299eb863d49ba0954b70388eb46f799268502>.

- [51] Chunzhi Li, Jiali Liu, He Li, Kaifeng Wu, Junhui Wang, and Qihua Yang. Covalent organic frameworks with high quantum efficiency in sacrificial photocatalytic hydrogen evolution. In *Nature Communications*, 2022. doi: 10.1038/s41467-022-30035-x. URL <https://doi.org/10.1038/s41467-022-30035-x>.
- [52] Panjie Li, Linsen Hao, Siyu Yang, Jie Yang, Siqi Tong, and Jiang Jing. Progressive enhancement of core π -conjugation of covalent organic frameworks to promote exciton dissociation for high-efficiency photosynthesis of H_2O_2 . *Journal of Environmental Chemical Engineering*, 2026. doi: 10.1016/j.jece.2026.124373. URL <https://www.semanticscholar.org/paper/cb8a41c346c12b1a75c15d348de4667a3fdde8fd>.
- [53] Zhiqing Lin, Songyao Dai, Shan Yao, Qia-Chun Lin, Mengying Fu, Lai-Hon Chung, Bin Han, and Jun He. Diacetylene-bridged covalent organic framework as crystalline graphdiyne analogue for photocatalytic hydrogen evolution. In *Chemical Science*, 2024. doi: 10.1039/d4sc06633b. URL <https://doi.org/10.1039/d4sc06633b>.
- [54] Cheng Liu, Yumei Xiao, Qing Yang, Yongchao Wang, Ruowei Lu, Yanxia Chen, Cuijuan Wang, and Hongjian Yan. A highly fluorine-functionalized 2D covalent organic framework for promoting photocatalytic hydrogen evolution, 2021. URL <https://www.semanticscholar.org/paper/542dee2f551f32f09c786989ada538c44ec6e35f>.
- [55] Nengyi Liu, Shuailei Xie, Yuxing Huang, Jiaping Lu, Hongjie Shi, Shumeng Xu, Guigang Zhang, and Xiong Chen. Dual-acceptor engineering in pyrene-based covalent organic frameworks for boosting photocatalytic hydrogen evolution. In *Advanced Energy Materials*, 2024. doi: 10.1002/aenm.202402395. URL <https://doi.org/10.1002/aenm.202402395>.
- [56] Ning Liu, Junzhou He, and Jingwei Liu. Donor-acceptor type covalent organic frameworks for photocatalytic hydrogen evolution. In *ChemistrySelect*, 2025. doi: 10.1002/slct.202405672. URL <https://doi.org/10.1002/slct.202405672>.
- [57] Zhenkun Liu, Ya Zhang, Youlin Wu, Bolin Yang, Zhengyu Zhou, and Zhiliang Jin. In situ XPS evidence of fully conjugated COF and C_3N_4 construct S-scheme heterojunction boosting photogenerated carriers transfer and separate for efficiently photocatalytic hydrogen evolution. *Journal of Material Science and Technology*, 2025. doi: 10.1016/j.jmst.2025.01.040. URL <https://doi.org/10.1016/j.jmst.2025.01.040>.
- [58] Yang Lu, Wenyan Li, Chenyu Sun, Yawen Tang, Lei Cheng, and Hanjun Sun. Copper-surface-mediated synthesis of sp^2 carbon-conjugated covalent organic framework photocathodes for photoelectrochemical hydrogen evolution. *Chemistry - A European Journal*, 2024. doi: 10.1002/chem.202402930. URL <https://doi.org/10.1002/chem.202402930>.
- [59] Xiao Luo, Yunlei Wang, Haifeng Lv, and Xiaojun Wu. Asymmetric potential model of two-dimensional imine covalent organic frameworks with enhanced quantum efficiency for photocatalytic water splitting. *Journal of Physical Chemistry Letters*, 2024. doi: 10.1021/acs.jpclett.4c00980. URL <https://www.semanticscholar.org/paper/6558228dd64119483d92d2728bf244ccfe10be74>.
- [60] Alberto López-Magano, Saba Daliran, Ali Reza Oveisi, Rubén Mas-Ballesté, Amarajothi Dhakshinamoorthy, José Alemán, Hermenegildo García, and Rafael Luque. Recent advances in the use of covalent organic frameworks as heterogenous photocatalysts in organic synthesis. In *Advanced Materials*, 2022. doi: 10.1002/adma.202209475. URL <https://doi.org/10.1002/adma.202209475>.

- [61] Fangpei Ma, Xiao Chi, Ying Wen, Qihong Yue, Tao Chen, Xiaojiang Yu, Xiaoling Liu, Yu Zhou, and Jun Wang. Engineering charge transfer by tethering halogens to covalent organic frameworks for photocatalytic sacrificial hydrogen evolution. In *Chemical Science*, 2025. doi: 10.1039/d5sc00082c. URL <https://doi.org/10.1039/d5sc00082c>.
- [62] Jiani Ma, Tina Jingyan Miao, and Junwang Tang. Charge carrier dynamics and reaction intermediates in heterogeneous photocatalysis by time-resolved spectroscopies. In *Chemical Society Reviews*, 2022. doi: 10.1039/d1cs01164b. URL <https://doi.org/10.1039/d1cs01164b>.
- [63] Si Ma, Zhongping Li, Yuxin Hou, Jiali Li, Zhenwei Zhang, Tianqi Deng, Gang Wu, Rui Wang, Shuo-wang Yang, and Xiaoming Liu. Fully conjugated benzobisoxazole-bridged covalent organic frameworks for boosting photocatalytic hydrogen evolution. In *Angewandte Chemie International Edition*, 2025. doi: 10.1002/anie.202501869. URL <https://doi.org/10.1002/anie.202501869>.
- [64] Tiantian Ma, Zhijie Liu, Jiaqi Deng, Enshan Han, Jun Liang, and Ruihu Wang. Ii-scheme heterojunction frameworks based on covalent organic frameworks and HKUST-1 for boosting photocatalytic hydrogen evolution. In *ChemSusChem*, 2024. doi: 10.1002/cssc.202400987. URL <https://www.semanticscholar.org/paper/ea396892d849ad577a8e8dd85994ecde828afbc2>.
- [65] Yuting Ma, Shenhui Yu, Wei Li, Di Chen, Zhenqian Zheng, Linjie Mao, Xuan Yang, Wenjun Wang, and Pingwei Liu. Rapid yet controlled synthesis of 2D covalent organic framework nanocapsules as high-performance photocatalytic carriers. In *Angewandte Chemie International Edition*, 2024. doi: 10.1002/anie.202416980. URL <https://doi.org/10.1002/anie.202416980>.
- [66] Z. Mi, Ting Zhou, Weijun Weng, Junjuda Unruangsri, Ke Hu, Wuli Yang, Changchun Wang, Kai A. I. Zhang, and Jia Guo. Covalent organic frameworks enabling site-isolation of viologen-derived electron transfer mediators for stable photocatalytic hydrogen evolution. In *Angewandte Chemie*, 2021. doi: 10.1002/anie.202016618. URL <https://www.semanticscholar.org/paper/b3477e018c3dacda249f9e6ae9c02685fa29ddf9>.
- [67] Bikash Mishra, Akhtar Alam, Bidhan Kumbhakar, David Díaz Díaz, and Pradip Pachfule. Impact of the crystallinity of covalent organic frameworks on photocatalytic hydrogen evolution. In *Crystal Growth & Design*, 2023. doi: 10.1021/acs.cgd.3c00379. URL <https://doi.org/10.1021/acs.cgd.3c00379>.
- [68] Bikash Mishra, Akhtar Alam, Avanti Chakraborty, Bidhan Kumbhakar, Samrat Ghosh, Pradip Pachfule, and Arne Thomas. Covalent organic frameworks for photocatalysis. In *Advanced Materials*, 2025. doi: 10.1002/adma.202413118. URL <https://doi.org/10.1002/adma.202413118>.
- [69] Chunshao Mo, Meijia Yang, Fusai Sun, Junhua Jian, Linfeng Zhong, Zhengsong Fang, Jiangshan Feng, and Dingshan Yu. Alkene-linked covalent organic frameworks boosting photocatalytic hydrogen evolution by efficient charge separation and transfer in the presence of sacrificial electron donors. In *Advanced Science*, 2020. doi: 10.1002/advs.201902988. URL <https://doi.org/10.1002/advs.201902988>.
- [70] Amit Nagar, Akhtar Alam, Pradip Pachfule, and C. M. Nagaraja. Isomeric cyano-vinylene-linked covalent organic frameworks and their impact on photocatalytic hydrogen evolution. *Journal of Materials Chemistry A*, 2025. doi: 10.1039/d5ta01539a. URL <https://doi.org/10.1039/d5ta01539a>.

- [71] Syeda Andleeb Zahra Naqvi, Ateeq Ur Rehman Baloch, Banothu Rammurthy, Jinsong Chen, Yan-Xi Tan, and Yaobing Wang. Design and synthesis of trithiophene-based donor-acceptor covalent organic framework for cocatalyst-free photocatalytic hydrogen evolution. In *ChemSusChem*, 2026. doi: 10.1002/cssc.202502428. URL <https://doi.org/10.1002/cssc.202502428>.
- [72] Qing Niu, Linhua Mi, Wei Chen, Qiujuan Li, Shenghong Zhong, Yan Yu, and Liuyi Li. Review of covalent organic frameworks for single-site photocatalysis and electrocatalysis. *Chinese Journal of Catalysis*, 2023. doi: 10.1016/s1872-2067(23)64457-2. URL [https://doi.org/10.1016/s1872-2067\(23\)64457-2](https://doi.org/10.1016/s1872-2067(23)64457-2).
- [73] Yuanyuan Niu, Feng Liu, Ping Li, Hongbin Qi, and Bing Sun. Fluorine-substituted covalent organic framework/anodized TiO₂ z-scheme heterojunction for enhanced photoelectrochemical hydrogen evolution. In *Catalysts*, 2026. doi: 10.3390/catal16010108. URL <https://www.semanticscholar.org/paper/5b26317ec086b4867a04aec2e15cca1ffc08eb73>.
- [74] Martin Osemba, Loice Ojwang, Adrián Chávez Huerta, and Godfrey Keru. Synergistic dual single-atom Fe–Co catalysts anchored on covalent organic framework/g-C₃N₄ heterostructures for ultrabroadband artificial photosynthesis via solar co₂ reduction and hydrogen evolution, 2026. URL <https://doi.org/10.26434/chemrxiv.15005609/v1>.
- [75] Ailing Pan, Xiaohui Sun, Yuanyuan Che, Yu Wang, and Hong Du. Reducing the excitation binding energy of covalent-organic frameworks via spatial-confined NiCo-NC as internal nanoreactors for photocatalytic hydrogen evolution. *Journal of Colloid and Interface Science*, 2025. doi: 10.1016/j.jcis.2025.137944. URL <https://www.semanticscholar.org/paper/c56278b2964fd3e7a419d84530a5c5d2f13d22d7>.
- [76] Yimin Pan, Yifan Dong, Zhenze Yang, Ailin Pan, Wenjie Shi, Wenbin Lin, and Haifeng Zheng. Rational synthesis of photocatalytic acridinium-based covalent organic frameworks via single-atom skeletal editing. In *Angewandte Chemie International Edition*, 2025. doi: 10.1002/anie.202514572. URL <https://doi.org/10.1002/anie.202514572>.
- [77] Shiqiong Peng, Renjie Wang, Yao Yang, Shuyan Wang, En Liang, Bing Han, Junbo Li, Xianglin Yu, and Qichun Zhang. sp² carbon-conjugated covalent organic frameworks (sp²c-COFs): Synthesis and application in photocatalytic water splitting. In *Macromolecular rapid communications*, 2025. doi: 10.1002/marc.202400967. URL <https://www.semanticscholar.org/paper/4e6ad7518f7fd78399d7cec283845c26032949e8>.
- [78] Lanyue Qi, Chengming Xiao, Wenyuan Lu, Hao Zhang, Yujun Zhou, Junwen Qi, Yue Yang, Zhigao Zhu, and Jiansheng Li. Triazine-based covalent organic framework/ g-C₃N₄ heterojunction toward highly efficient photoactivation of peroxydisulfate for sulfonamides degradation. In *Separation and Purification Technology*, 2024. doi: 10.1016/j.seppur.2024.128758. URL <https://www.semanticscholar.org/paper/31bb69015af16ab7d356af0f598b6f1f9f13df1e>.
- [79] Qi Qi, Jiawen Liu, Hang Xu, Wei Wen, and Zhonghua Li. Construction of covalently and coordination-linked MOF@COF heterojunctions via interfacial engineering for efficient photocatalytic hydrogen evolution. *International Journal of Hydrogen Energy*, 2026. doi: 10.1016/j.ijhydene.2026.156278. URL <https://doi.org/10.1016/j.ijhydene.2026.156278>.
- [80] Shi-peng Qi, Rui-tang Guo, Zhe-xu Bi, Zhen-rui Zhang, Chu-fan Li, and Weiguo Pan. Recent progress of covalent organic frameworks-based materials in photocatalytic applications: A review. In *Small*, 2023. doi: 10.1002/sml.202303632. URL <https://doi.org/10.1002/sml.202303632>.

- [81] Yuniang Qian, Yulan Han, Xiyuan Zhang, Ge Yang, Guozhen Zhang, and Hai-Long Jiang. Computation-based regulation of excitonic effects in donor-acceptor covalent organic frameworks for enhanced photocatalysis. In *Nature Communications*, 2023. doi: 10.1038/s41467-023-38884-w. URL <https://doi.org/10.1038/s41467-023-38884-w>.
- [82] Liyang Qin, Chengdi Ma, Jian Zhang, and Tianhua Zhou. Structural motifs in covalent organic frameworks for photocatalysis. In *Advanced Functional Materials*, 2024. doi: 10.1002/adfm.202401562. URL <https://doi.org/10.1002/adfm.202401562>.
- [83] Zhishi Qiu, Zhongge Luo, Chunting Zhao, Tong Zhou, Mingpeng Chen, Baoye Zi, Ke Wang, Jin Zhang, Jianhong Zhao, Yumin Zhang, and Qingju Liu. Atomically anchored Pt-n sites as bidirectional electron highways in covalent organic frameworks for the synergistic enhancement of photocatalytic hydrogen evolution. In *Science Bulletin*, 2026. doi: 10.1016/j.scib.2026.03.025. URL <https://www.semanticscholar.org/paper/c157d1341a66b63134ebbf69bc1eb6a96242047>.
- [84] Konggang Qu, Zhifei Chen, Lihui Wang, Haibo Li, Suyuan Zeng, Rui Li, Lijian Meng, Hongyan Chen, and Qingxia Yao. Covalent organic framework assisted low-content ultrafine Ru on porous n-doped carbon for efficient hydrogen evolution reaction. In *Rare Metals*, 2024. doi: 10.1007/s12598-024-03024-w. URL <https://doi.org/10.1007/s12598-024-03024-w>.
- [85] Huili Ran, Quanlong Xu, Yun Yang, Huaxing Li, Jiajie Fan, Gang Liu, Lijie Zhang, Jun Zou, Huile Jin, and Shun Wang. Progress of covalent organic framework photocatalysts: From crystallinity–stability dilemma to photocatalytic performance improvement. In *ACS Catalysis*, 2024. doi: 10.1021/acscatal.4c02738. URL <https://www.semanticscholar.org/paper/518ef724cc5bf90cb392b942b4e0fcef1608b7a2>.
- [86] Andrés Rodríguez-Camargo and B. Lotsch. A reflection on ‘a hydrazone-based covalent organic framework for photocatalytic hydrogen production’: teaching sponges new tricks. In *Chemical Science*, 2026. doi: 10.1039/d6sc90032a. URL <https://www.semanticscholar.org/paper/240b47b6c2ec8ed2257be9a913635e61b2aa307d>.
- [87] Nuria Romero, Roger Bofill, Laia Francàs, Jordi García-Antón, and Xavier Sala. Light-driven hydrogen evolution assisted by covalent organic frameworks. In *Catalysts*, 2021. doi: 10.3390/catal11060754. URL <https://doi.org/10.3390/catal11060754>.
- [88] Dandan Shang, Di Li, Biyi Chen, Bifu Luo, Yuanyong Huang, and Weidong Shi. 2D–2D sns 2 /covalent organic framework heterojunction photocatalysts for highly enhanced solar-driven hydrogen evolution without cocatalysts. In *ACS Sustainable Chemistry & Engineering*, 2021. doi: 10.1021/acssuschemeng.1c05162. URL <https://doi.org/10.1021/acssuschemeng.1c05162>.
- [89] Rongchen Shen, Xiuzhi Li, Chaochao Qin, Peng Zhang, and Xin Li. Efficient photocatalytic hydrogen evolution by modulating excitonic effects in Ni-intercalated covalent organic frameworks. In *Advanced Energy Materials*, 2023. doi: 10.1002/aenm.202203695. URL <https://doi.org/10.1002/aenm.202203695>.
- [90] Rongchen Shen, Guijie Liang, Lei Hao, Peng Zhang, and Xin Li. In situ synthesis of chemically bonded 2D/2D covalent organic frameworks/o-vacancy wo 3 z-scheme heterostructure for photocatalytic overall water splitting. In *Advanced Materials*, 2023. doi: 10.1002/adma.202303649. URL <https://doi.org/10.1002/adma.202303649>.

- [91] Rongchen Shen, Can Huang, Lei Hao, Guijie Liang, Peng Zhang, Qiang Yue, and Xin Li. Ground-state charge transfer in single-molecule junctions covalent organic frameworks for boosting photocatalytic hydrogen evolution. In *Nature Communications*, 2025. doi: 10.1038/s41467-025-57662-4. URL <https://doi.org/10.1038/s41467-025-57662-4>.
- [92] A. R. Sujatha, Aparna Anil, Princy Deni Raju, and C. Suneesh. 1,3,5-triformylphloroglucinol derived β -ketoenamine-linked functional covalent organic frameworks with enhanced crystallinity and stability-recent advances. *Chemistry - An Asian Journal*, 2025. doi: 10.1002/asia.202401434. URL <https://www.semanticscholar.org/paper/9b070f491cca0bf72072a5c18b26216166b135f4>.
- [93] Tao Sun, Rui Wang, Renquan Guan, Li Wang, Tian Zhong, Chunbo Liu, Xueying Cheng, and Qianrong Fang. Design strategies and application progress of covalent organic frameworks in photocatalytic oxidation reactions. In *Chemical Science*, 2025. doi: 10.1039/d5sc08410e. URL <https://doi.org/10.1039/d5sc08410e>.
- [94] Junwang Tang, James R. Durrant, and David R. Klug. Mechanism of photocatalytic water splitting in tio 2 . reaction of water with photoholes, importance of charge carrier dynamics, and evidence for four-hole chemistry. *Journal of the American Chemical Society*, 2008. doi: 10.1021/ja8034637. URL <https://doi.org/10.1021/ja8034637>.
- [95] Wenqi Tang, Jianrong Zeng, Fenggang Bian, Huiming Lin, Huanyu Zhao, and Zhongyuan Lv. Engineering heteroatomic structures in Pt single-atom anchored covalent organic frameworks for enhanced photocatalytic hydrogen evolution reaction, 2025. URL <https://doi.org/10.2139/ssrn.5118948>.
- [96] Yangyang Wan, Lei Wang, Hangxun Xu, Xiaojun Wu, and Jinlong Yang. A simple molecular design strategy for two-dimensional covalent organic framework capable of visible-light-driven water splitting. *Journal of the American Chemical Society*, 2020. doi: 10.1021/jacs.0c00564. URL <https://doi.org/10.1021/jacs.0c00564>.
- [97] Lijia Wang, Pengyu Dong, Guiling Zhang, and Fengming Zhang. Review and perspectives of β -keto-enamine-based covalent organic framework for photocatalytic hydrogen evolution. In *Energy & Fuels*, 2023. doi: 10.1021/acs.energyfuels.3c00162. URL <https://doi.org/10.1021/acs.energyfuels.3c00162>.
- [98] Ming Wang, Yaling Li, Dengxin Yan, Hui Hu, Yujie Song, Xiaofang Su, Jiamin Sun, Songtao Xiao, and Yanan Gao. Dipole polarization modulating of vinylene-linked covalent organic frameworks for efficient photocatalytic hydrogen evolution. *CHINESE JOURNAL OF CATALYSIS (CHINESE VERSION)*, 2024. doi: 10.1016/s1872-2067(24)60113-0. URL [https://doi.org/10.1016/s1872-2067\(24\)60113-0](https://doi.org/10.1016/s1872-2067(24)60113-0).
- [99] Penghua Wang, Chenghan Yang, Huan Liu, Huijun Yu, Tong Bian, Yuming Zhou, and Yiwei Zhang. Trinuclear copper-cluster-bridged covalent organic framework with synergistic zinc single atoms for photocatalytic hydrogen evolution. *European Journal of Inorganic Chemistry*, 2026. doi: 10.1002/ejic.202500565. URL <https://doi.org/10.1002/ejic.202500565>.
- [100] Shuo Wang, Y. Zeng, Miaomiao Fan, Juan Du, Jianlin Hao, and Zhengcheng Li. Bandgap-dependent Pt size on covalent organic frameworks: the COF structure outweighs the Pt cocatalyst size in photocatalytic hydrogen evolution. In *RSC Advances*, 2026. doi: 10.1039/d6ra04831e. URL <https://www.semanticscholar.org/paper/a1395a99e2bd6b4d0d802d6df22a83512ef487d9>.

- [101] Xiaoyan Wang, Linjiang Chen, Samantha Y. Chong, Marc A. Little, Yongzhen Wu, Weihong Zhu, Rob Clowes, Yong Yan, Martijn A. Zwijnenburg, Reiner Sebastian Sprick, and Andrew I. Cooper. Sulfone-containing covalent organic frameworks for photocatalytic hydrogen evolution from water. In *Nature Chemistry*, 2018. doi: 10.1038/s41557-018-0141-5. URL <https://doi.org/10.1038/s41557-018-0141-5>.
- [102] Yaqin Wang, Zelong Qiao, Han Li, Rui Zhang, Zhonghua Xiang, Dapeng Cao, and Shitao Wang. Molecular engineering for modulating photocatalytic hydrogen evolution of fully conjugated 3D covalent organic frameworks. In *Angewandte Chemie*, 2024. doi: 10.1002/anie.202404726. URL <https://www.semanticscholar.org/paper/2fb5f0de2bedfa25f09844429b4d3feb5cbf5063>.
- [103] Yu Wang, Ailing Pan, Yuhua Ma, and Hong Du. Single-atom Co anchored on covalent organic framework for remarkable photocatalytic hydrogen evolution efficiency. *International Journal of Hydrogen Energy*, 2024. doi: 10.1016/j.ijhydene.2024.05.448. URL <https://doi.org/10.1016/j.ijhydene.2024.05.448>.
- [104] Weijun Weng and Jia Guo. The effect of enantioselective chiral covalent organic frameworks and cysteine sacrificial donors on photocatalytic hydrogen evolution. In *Nature Communications*, 2022. doi: 10.1038/s41467-022-33501-8. URL <https://doi.org/10.1038/s41467-022-33501-8>.
- [105] Weijun Weng, Zheng Lin, HuaLei Zhang, Fushuang Niu, Changchun Wang, Ke Hu, and Jia Guo. Effect of ES IPT-induced photoisomerization of keto-enamine linkages on the photocatalytic hydrogen evolution performance of covalent organic frameworks. In *JACS Au*, 2023. doi: 10.1021/jacsau.3c00554. URL <https://doi.org/10.1021/jacsau.3c00554>.
- [106] Dongchuang Wu, Ning Gu, Junru Yao, Yang Cao, Lun Wang, I. Shakir, Youyi Sun, and Yuxi Xu. Recent advances in room-temperature synthesis of covalent organic frameworks. In *Chemical Science*, 2025. doi: 10.1039/d5sc00109a. URL <https://www.semanticscholar.org/paper/768bed5e8a8a710dc986dc8004748cd1cefff148>.
- [107] Xinhe Wu and Chang Cheng. Chiral COF-based S-scheme heterojunction for spin-polarized photocatalytic hydrogen evolution. In *Acta Physico-Chimica Sinica*, 2026. doi: 10.1016/j.actphy.2026.100303. URL <https://doi.org/10.1016/j.actphy.2026.100303>.
- [108] Yueyuan Xiao, Kuixing Wang, Wenbo Dong, and Longyu Li. Molecular engineering in tunable crystallinity of 2D covalent organic frameworks for efficient photocatalytic hydrogen evolution. In *Polymer*, 2024. doi: 10.1016/j.polymer.2024.126980. URL <https://doi.org/10.1016/j.polymer.2024.126980>.
- [109] Minghua Xu, Xiaowen Ruan, Xinlei Zhang, Jiaqi Guo, Chunsheng Ding, Dongxu Jiao, Depeng Meng, Guozhen Fang, Jing Leng, Kaikai Ba, Tengfeng Xie, Wei Zhang, Sai Kishore Ravi, Enquan Jin, and Xiaoqiang Cui. Localized oxygen enrichment in a covalent organic framework-ZnIn₂S₄ s-scheme heterojunction enables spatially confined oxygen reduction and boosts photocatalytic h₂o₂ selectivity. *Nano letters (Print)*, 2025. doi: 10.1021/acs.nanolett.5c03476. URL <https://www.semanticscholar.org/paper/ddf39b17a844a3dd5f649c96fe65440403ffe5ce>.
- [110] Naizhang Xu, Yubing Liu, Wenjuan Yang, Jie Tang, Bowei Cai, Qi Li, Jiawei Sun, Kaiqiang Wang, Bolian Xu, Qitao Zhang, and Yining Fan. 2D-2D heterojunctions of a covalent triazine framework with a triphenylphosphine-based covalent organic framework for efficient photocatalytic hydrogen evolution. In *ACS Applied Energy Materials*, 2020. doi: 10.1021/acsaem.0c02102. URL <https://doi.org/10.1021/acsaem.0c02102>.

- [111] Yizhuo Xu, Nikita Rog, Catherine Mollart, Ganna Gryn'ova, Abbie Trewin, and Cher Hon Lau. Structural modulation of TpPa-1 covalent organic framework in flow synthesis guided by green chemistry principles. In *ACS Engineering Au*, 2026. doi: 10.1021/acsengineeringau.5c00114. URL <https://doi.org/10.1021/acsengineeringau.5c00114>.
- [112] Ping Xue, Mingyuan Li, Mi Tang, Zhengbang Wang, and Chengliang Wang. Research progress of β -ketoamine-linked covalent organic frameworks for photocatalytic hydrogen evolution. In *Organic Materials*, 2024. doi: 10.1055/a-2291-8578. URL <https://doi.org/10.1055/a-2291-8578>.
- [113] Ming Yan, Fan Jiang, and Yilin Wu. Metal-free 2D-2D black phosphorus/covalent organic framework p-n heterojunction for efficient visible-light-driven hydrogen evolution without cocatalysts. *International Journal of Hydrogen Energy*, 2023. doi: 10.1016/j.ijhydene.2022.12.002. URL <https://doi.org/10.1016/j.ijhydene.2022.12.002>.
- [114] Jin Yang, Amitava Acharjya, Mengyang Ye, Jabor Rabeah, Shuang Li, Zdravko Kochovski, Sol Youk, Jérôme Roeser, Julia Grüneberg, Christopher Penschke, Michael Schwarze, Tianyi Wang, Yan Lü, Roel van de Krol, Martin Oschatz, Reinhard Schomäcker, Peter Saalfrank, and Arne Thomas. Protonated imine-linked covalent organic frameworks for photocatalytic hydrogen evolution. In *Angewandte Chemie International Edition*, 2021. doi: 10.1002/anie.202104870. URL <https://doi.org/10.1002/anie.202104870>.
- [115] Jinglun Yang, Zihao Chen, Lei Zhang, and Qichun Zhang. Covalent organic frameworks for photocatalytic reduction of carbon dioxide: A review. In *ACS Nano*, 2024. doi: 10.1021/acsnano.4c06783. URL <https://www.semanticscholar.org/paper/aef40464ffb077310da4e99734aab9354809d8ab>.
- [116] Qing Yang, Maolan Luo, Kewei Liu, Hongmei Cao, and Hongjian Yan. Covalent organic frameworks for photocatalytic applications. In *Applied Catalysis B: Environmental*, 2020. doi: 10.1016/j.apcatb.2020.119174. URL <https://doi.org/10.1016/j.apcatb.2020.119174>.
- [117] Tian Yang, Wenjuan Li, Jianmei Yuan, Jia Wang, Jingjing Zhang, and Shujun Chao. Missing-linker defects in a porphyrin-based donor-acceptor covalent organic framework toward improved oxygen reduction, hydrogen evolution and dopamine sensing performance. *International Journal of Hydrogen Energy*, 2026. doi: 10.1016/j.ijhydene.2026.155289. URL <https://doi.org/10.1016/j.ijhydene.2026.155289>.
- [118] Yan Yang, Xiaoyu Chu, Hong-Yu Zhang, Rui Zhang, Yu-Han Liu, Feng-Ming Zhang, Meng Lu, Zhao-Di Yang, and Ya-Qian Lan. Engineering β -ketoamine covalent organic frameworks for photocatalytic overall water splitting. In *Nature Communications*, 2023. doi: 10.1038/s41467-023-36338-x. URL <https://doi.org/10.1038/s41467-023-36338-x>.
- [119] Jungang Yi, Linfeng Zhang, Wei Wang, Qun Yi, Huadong Wu, Jianding Li, and Jia Guo. Constructing the covalent organic framework and in2o3 composites via covalent bonds towards excellent visible-light photocatalytic hydrogen evolution. In *Fuel*, 2024. doi: 10.1016/j.fuel.2023.129470. URL <https://doi.org/10.1016/j.fuel.2023.129470>.
- [120] Junhua You, Yao Zhao, Lu Wang, and Wanting Bao. Recent developments in the photocatalytic applications of covalent organic frameworks: A review. *Journal of Cleaner Production*, 2021. doi: 10.1016/j.jclepro.2021.125822. URL <https://doi.org/10.1016/j.jclepro.2021.125822>.

- [121] Maria Younas, Muhammad Yar, Hamad AlMohamadi, Tariq Mahmood, Khurshid Ayub, Asim Laeeq Khan, Muhammad Yasin, and Mazhar Amjad Gilani. A rational design of covalent organic framework supported single atom catalysts for hydrogen evolution reaction: A DFT study. *International Journal of Hydrogen Energy*, 2024. doi: 10.1016/j.ijhydene.2023.07.062. URL <https://doi.org/10.1016/j.ijhydene.2023.07.062>.
- [122] Junjie Yu, Junjie Yu, Xin Huang, Liying Wang, Yanze Wu, Zhiwei Huang, L.Y. Su, Nannan Wang, Ji-Pan Yu, Ji-Pan Yu, and Wei-Qun Shi. Topology-templated synthesis of dibenzo[g,p]chrysene-based sp² carbon-linked covalent organic frameworks with Kagome lattice for enhanced photocatalytic hydrogen evolution. In *Advanced Materials*, 2025. doi: 10.1002/adma.202504808. URL <https://doi.org/10.1002/adma.202504808>.
- [123] Xinxin Yu, Yuzhen Zhao, Qiuyi Bao, Weina Wang, Yongpeng Li, Juanxiu Xiao, Zhuyin Sui, Xinlong Tian, and Qi Chen. Vinyl-bearing sp² carbon-conjugated covalent organic framework composites for enhanced electrochemical performance in hydrogen evolution reaction and lithium-sulfur batteries. *Journal of Colloid and Interface Science*, 2024. doi: 10.1016/j.jcis.2024.07.076. URL <https://www.semanticscholar.org/paper/680c1089c30b91c386aa6aab5e0db2359162ee2c>.
- [124] Song Yupeng, Hua Zihui, Ge Jiechao, Wang Chunru, and Wu Bo. C₆₀@TpPa covalent organic framework with strong built-in electric field enables efficient photocatalytic hydrogen production. In *Acta Chimica Sinica*, 2026. doi: 10.6023/a25100352. URL <https://doi.org/10.6023/a25100352>.
- [125] Fulin Zhang, Xiaoyun Dong, Yuexin Wang, and Xianjun Lang. Design and synthesis of a triazine-based sp² carbon-conjugated covalent organic framework for blue light photocatalysis. In *Small*, 2023. doi: 10.1002/smll.202302456. URL <https://doi.org/10.1002/smll.202302456>.
- [126] Fushuai Zhang, Rongjian Sa, Ke Kong, Zhibin Yang, Maosong Li, Yong-Min Lee, Ruihu Wang, and Wonwoo Nam. Electronic configurations modulated by heterometallic orbital interactions of spinel cocatalysts in covalent organic framework-based dyads for photocatalytic hydrogen evolution. In *ACS Catalysis*, 2026. doi: 10.1021/acscatal.5c08771. URL <https://www.semanticscholar.org/paper/4542cd0f61545b019379bcc36d9fd895259ed7d6>.
- [127] Hualei Zhang, Zheng Lin, Pinit Kidkhunthod, and Jia Guo. Stable immobilization of nickel ions on covalent organic frameworks for panchromatic photocatalytic hydrogen evolution. In *Angewandte Chemie International Edition*, 2023. doi: 10.1002/anie.202217527. URL <https://doi.org/10.1002/anie.202217527>.
- [128] Jie Zhang, Xuejing Li, Jianhong Chang, Ping She, Hui Li, Shilun Qiu, and Qianrong Fang. Dual electric fields in cyclooctatetrathiophene-based COF/znIn₂S₄ z-scheme heterojunction boost photocatalytic seawater hydrogen evolution. In *Angewandte Chemie International Edition*, 2025. doi: 10.1002/anie.202519752. URL <https://doi.org/10.1002/anie.202519752>.
- [129] Jie Zhang, Xuejing Li, Jianhong Chang, Ping She, Hui Li, Shilun Qiu, and Qianrong Fang. Dual electric fields in cyclooctatetrathiophene-based COF/znIn₂S₄ z-scheme heterojunction boost photocatalytic seawater hydrogen evolution. In *Angewandte Chemie*, 2025. doi: 10.1002/ange.202519752. URL <https://doi.org/10.1002/ange.202519752>.
- [130] Jingya Zhang, Xinyu Li, Qing Yang, Chunyan Tang, Kewei Liu, Zhishan Su, and Hongjian Yan. Bifunctional platinum cocatalysts with directional loading on a covalent organic framework for photocatalytic overall water splitting. In *Small*, 2026. doi: 10.1002/smll.75228. URL <https://www.semanticscholar.org/paper/51fd3208b770d6ed902a4af6a5d9638500b8d920>.

- [131] Xiangyu Zhang, Xiao Wu, Rufan Chen, and Qing-Hua Xu. A triazine-based covalent organic framework decorated with cadmium sulfide for efficient photocatalytic hydrogen evolution from water. *Journal of Colloid and Interface Science*, 2024. doi: 10.1016/j.jcis.2024.03.104. URL <https://doi.org/10.1016/j.jcis.2024.03.104>.
- [132] Yang-Peng Zhang, wei Han, yan yang, Hong-Yu Zhang, Ya Wang, Long Wang, Xiao-Jun Sun, and Feng-Ming Zhang. S-scheme heterojunction of black tio₂ and covalent-organic framework for enhanced photocatalytic hydrogen evolution, 2022. URL <https://doi.org/10.2139/ssrn.4052326>.
- [133] Yunchao Zhang, Jinkang Pan, Xiang Ni, Feiqi Mo, Yuanguo Xu, and Pengyu Dong. Revealing the dynamics of charge carriers in organic/inorganic hybrid fs-COF/wo₃ S-scheme heterojunction for boosted photocatalytic hydrogen evolution. *Chinese Journal of Catalysis*, 2025. doi: 10.1016/s1872-2067(25)64664-x. URL [https://doi.org/10.1016/s1872-2067\(25\)64664-x](https://doi.org/10.1016/s1872-2067(25)64664-x).
- [134] Chunting Zhao, Ran Shi, Xingyan Yang, Jianhong Zhao, Shan Zeng, Zhishi Qiu, Yumin Zhang, and Qingju Liu. Cocatalyst-free hydrogen evolution over benzothiazole-based metal covalent organic framework under visible light. *Chemical Engineering Journal*, 2025. doi: 10.1016/j.cej.2025.170127. URL <https://doi.org/10.1016/j.cej.2025.170127>.
- [135] Huihui Zhao, Xinghao Zhang, Chengcheng Yu, Wenya Gao, Xiuxiu Chen, Haikuo Lan, Xiaorong Xin, Kang Liu, and Dingxuan Ma. Covalent organic framework-derived highly dispersed Pt single atoms collaborate with Pt nanoclusters electrocatalyst for acid hydrogen evolution. In *Green Chemistry*, 2025. doi: 10.1039/d4gc05667a. URL <https://doi.org/10.1039/d4gc05667a>.
- [136] Wei Zhao, Liang Luo, Muyu Cong, Xueyan Liu, Zhiyun Zhang, Mounib Bahri, Boyu Li, Jing Yang, Miaojie Yu, Lunjie Liu, Yu Xia, Nigel D. Browning, Wei-Hong Zhu, Weiwei Zhang, and Andrew I. Cooper. Nanoscale covalent organic frameworks for enhanced photocatalytic hydrogen production. In *Nature Communications*, 2024. doi: 10.1038/s41467-024-50839-3. URL <https://doi.org/10.1038/s41467-024-50839-3>.
- [137] Yanyan Zhao, Chunyan Yang, Shumin Zhang, Guotai Sun, Bicheng Zhu, Linxi Wang, and Jianjun Zhang. Investigating the charge transfer mechanism of ZnSe qd/COF S-scheme photocatalyst for h₂o₂ production by using femtosecond transient absorption spectroscopy. *Chinese Journal of Catalysis*, 2024. doi: 10.1016/s1872-2067(24)60069-0. URL [https://doi.org/10.1016/s1872-2067\(24\)60069-0](https://doi.org/10.1016/s1872-2067(24)60069-0).
- [138] Yuxiang Zhao, Ying Liang, Daoxiong Wu, Hao Tian, Tian Xia, Wenxin Wang, Weiyu Xie, Xin-Ming Hu, Xinlong Tian, and Qi Chen. Ruthenium complex of sp² carbon-conjugated covalent organic frameworks as an efficient electrocatalyst for hydrogen evolution. In *Small*, 2022. doi: 10.1002/smll.202107750. URL <https://doi.org/10.1002/smll.202107750>.
- [139] Yuelin Zhong, Wenbo Dong, Shijie Ren, and Longyu Li. Oligo(phenylenevinylene)-based covalent organic frameworks with Kagome lattice for boosting photocatalytic hydrogen evolution. In *Advanced Materials*, 2023. doi: 10.1002/adma.202308251. URL <https://doi.org/10.1002/adma.202308251>.
- [140] Ting Zhou, Lei Wang, Xingye Huang, Junjuda Unruangsri, Hualei Zhang, Rong Wang, Qingliang Song, Qingyuan Yang, Weihua Li, Changchun Wang, Kaito Takahashi, Hangxun Xu, and Jia Guo. PEG-stabilized coaxial stacking of two-dimensional covalent organic frameworks for enhanced photocatalytic hydrogen evolution. In *Nature Communications*, 2021. doi: 10.1038/s41467-021-24179-5. URL <https://doi.org/10.1038/s41467-021-24179-5>.

- [141] Yijun Zhou, Pengyu Dong, Jinhong Liu, Beibei Zhang, Boyuan Zhang, Xinguo Xi, and Jinlong Zhang. Functional groups-dependent Tp-based COF/mgIn₂S₄ s-scheme heterojunction for photocatalytic hydrogen evolution. In *Advanced Functional Materials*, 2025. doi: 10.1002/adfm.202500733. URL <https://doi.org/10.1002/adfm.202500733>.
- [142] Zilong Zhou, Yuting Xiao, Jing Tian, Ning Nan, Ren-Jie Song, and Jin-Heng Li. Recent advances in metal-free covalent organic frameworks for photocatalytic applications in energy and environmental fields. *Journal of Materials Chemistry A*, 2023. doi: 10.1039/d2ta09582c. URL <https://doi.org/10.1039/d2ta09582c>.
- [143] Shipeng Zhu, Zhipeng Luo, Fengtao Zhang, Yu Huang, Wanxiang Yang, Yujing Xiong, Ruiying Zhang, Ziyue Yu, Chao Lin, Xinyi Zeng, Shumeng Xu, Yuanming Li, Wei Lin, Jiayi Chen, and Xiong Chen. Fluorination-modulated molecular engineering in β -ketoenamine covalent organic frameworks toward efficient photocatalytic hydrogen evolution †. *Chinese Journal of Chemistry*, 2025. doi: 10.1002/cjoc.70349. URL <https://doi.org/10.1002/cjoc.70349>.